



Transorgan net fluxes of essential amino acids and their calculated efficiency of utilization in dairy cows: A meta-analysis

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ABSTRACT

The 8th revision of the Nutrient Requirements of Dairy Cattle by the National Academies of Sciences, Engineering, and Medicine (NASEM) introduced the concept of target efficiency, based on the calculation of the efficiency of utilization (EffU) of EAA. This EffU calculated using the NASEM approach (NASEM_{EffU}) represents the sum of each EAA in protein secretions and accretions divided by the metabolizable supply of each EAA, excluding the EAA contribution to endogenous urinary losses (assumed to have an EffU of 100%). Transorgan net fluxes across the portal-drained viscera (PDV), the liver, and the mammary gland were used to estimate the potential catabolism of EAA across these tissues and summed to yield the whole-body EAA catabolism from which inefficiency of use could be estimated. Therefore, EffU was calculated and referred to as Measured_{EffU}. Those measurements based on transorgan net fluxes give us the opportunity to verify how the theoretical NASEM_{EffU} compares with Measured_{EffU} in lactating dairy cows, which was the objective of the current meta-analysis. The dataset needed to be divided between studies where postsplanchnic supply of EAA was insufficient to account for mammary uptake and studies where it was adequate, resulting in one category for cows with DIM ≤30, and another for cows with DIM >30, referred to as cows in early lactation and later lactation, respectively. For cows in later lactation, NASEM_{EffU} and Measured_{EffU} agreed well (concordance correlation coefficient averaging 68%) based on significant catabolism of (1) group 1 AA (His, Met, and Phe) in the liver; (2) Ile, Leu and Val by the PDV, nonmammary peripheral tissues, and the mammary gland; (3) Lys in the liver, nonmammary peripheral tissues, and the mammary gland; and (4) Thr in the PDV, liver and the mammary gland. There was,

however, a significant mean bias between NASEM_{EffU} and Measured_{EffU} for Met, the latter being smaller than NASEM_{EffU}. This would suggest that the part of Met liver removal is used for the synthesis of metabolites, such as cysteine, glutathione, and taurine, not needing other EAA for their synthesis and not included into export proteins. This might suggest that the target EffU of Met proposed by NASEM (2021) should be reduced. For cows in early lactation, the concept of NASEM_{EffU} does not apply for most of the EAA. Mathematically, relationships between NASEM_{EffU} and Measured_{EffU} were poor in these animals, with the concordance correlation coefficient smaller than 30% for His, Met, Phe, Lys, and Val. Indeed, unknown endogenous supply from body protein mobilization invalidates the quantification of the actual contribution supporting protein export. The sites of EAA removal, however, remained the same throughout lactation except for the nonmammary peripheral tissues being a source of endogenous supply for all EAA, rather than a site for the catabolism for Ile, Leu, Lys, and Val, as observed for cows in later lactation.

Key words: liver, mammary gland, peripheral tissues, portal-drained viscera

INTRODUCTION

Balancing dietary protein to the lactating dairy cow should aim to optimize quantitatively the individual EAA supply required to support productive and nonproductive functions, such as milk, maintenance, health, and reproduction. In the National Academies of Science, Engineering, and Medicine's (NASEM) eighth revised edition of the *Nutrient Requirements of Dairy Cattle* (NASEM, 2021), a calculation of the efficiency (Eff) of utilization (EffU) of metabolizable EAA (mEAA) has been proposed for each of the 9 EAA. This efficiency represents the sum of AA exported as protein secretions and accretions scaled to the supply of mEAA. The AA exported excludes AA in endogenous urinary losses (EndoUri) and used for gestation, and mEAA discounts the AA

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

contributing to EndoUri, because the EffU were assumed to be fixed at 100% for EndoUri and 33% for gestation. Acknowledging the concept of a variable EffU of mEAA to support protein export, NASEM (2021) established recommendations of mEAA based on a target EffU for each EAA. These target EffU, assumed to be the same for protein secretion and accretion functions, vary from 60% for Phe to 86% for Trp. Conceptually, the amount of mEAA not used to support secretion and accretion would be catabolized, but direct measures of EAA catabolism in dairy cows have been restricted to Leu (e.g., Bequette et al., 1996; Lapierre et al., 2002; Martineau et al., 2007). Nonetheless, adequate predictions of mEAA by NASEM (Martineau et al., 2024, 2025; Martins et al., 2025) and accumulated data on transorgan net fluxes of EAA from several studies could help quantify their potential net catabolism, defined as the net flux of EAA across a tissue not contributing to protein export.

Initially, the EAA were classified into 2 groups with respect to their mammary utilization (Mephram, 1982), coherent with later measurements of their utilization by the splanchnic and postsplanchnic tissues (i.e., non-mammary peripheral tissues [PERI] + mammary gland; Lapierre et al., 2012). In the mammary gland, each EAA from the initial group 1 AA (His, Met, Phe+Tyr, and Trp) presented an apparent stoichiometric transfer to milk protein secretion synthesized in the mammary gland (i.e., excluding blood-borne proteins [MilkAA_{MG}]) on a net basis (Mephram, 1982; Lapierre et al., 2012). This results in a mammary uptake of AA to MilkAA_{MG} output ratio (U:O) close to 1, indicating little or no mammary catabolism. Furthermore, transorgan net flux data indicated that group 1 AA catabolism occurred mainly in the liver (Lapierre et al., 2005; Lapierre et al., 2012) with little or no potential catabolism by the portal-drained viscera (PDV; Martineau et al., 2025). In contrast, group 2 AA (the branched-chain AA [BCAA], Lys, and Thr) are taken up in excess relative to their MilkAA_{MG} (Mephram, 1982; Lapierre et al., 2012). The associated excess N and carbon relative to MilkAA_{MG} output increases with incremental mEAA supply and can be used for de novo synthesis of NEAA and intermediates of the tricarboxylic acid cycle in mammary cells (Mephram, 1982; Lapierre et al., 2009; Lapierre et al., 2012; NASEM, 2021). In mammary explants from lactating cows, Bequette et al. (2006) suggested that as much as 12% of galactose synthesis might derive from EAA catabolism. Overall, the BCAA and Lys are preferentially oxidized in extrahepatic tissues (Lapierre et al., 2002; Martineau et al., 2025) and barely removed, on a net basis, by the liver (Raggio et al., 2004; Berthiaume et al., 2006; Lapierre et al., 2012).

Therefore, the EffU of mEAA were estimated either based on the approach proposed by NASEM (NASEM_{EffU}), which involves NASEM-based export proteins

or using measurements of splanchnic and mammary net fluxes of EAA to estimate potential catabolism in different tissues (Measured_{EffU}). The hypothesis was that the potential catabolism, that is, the fractions of mEAA that were removed across the various tissues but could not be ascribed to export protein production (as defined in NASEM), would represent the inefficiency of utilization of EAA. Consequently, the objective of this meta-analysis was to investigate the association between NASEM_{EffU} and Measured_{EffU}. Such information may also assist future research in identifying key regulatory steps supporting milk true protein yield (MPY).

MATERIALS AND METHODS

Because no human or animal subjects were used, this analysis did not require approval by an Institutional Animal Care and Use Committee or Institutional Review Board. All data handling and statistical analyses were completed using R (version 4.5.2; R Core Team, 2025).

Data Collection

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (Moher et al., 2009) were followed in this study and the key words used for the search query are reported in Figure 1. A comprehensive literature search was conducted to identify studies reporting PDV net flux (PDV_{NF}) and splanchnic tissues net flux (TSP_{NF}) of one or more EAA in lactating dairy cows with DIM ranging from 1 to 305 d. The literature search was performed between April 2022 and September 2024, and included 2 search engines: the Scopus database (<https://www.scopus.com>) and Google Scholar (<https://scholar.google.com/>). Studies to be included in the current meta-analysis were required to report DMI, CP intake, feed ingredients and their inclusion rates.

The dataset did not include treatments where rumen-protected EAA were fed, although values from Dalbach et al. (2011) were included because the contribution of 2-hydroxy-4-(methylthio)-butanoic acid isopropyl ester (HMBi) to the PDV_{NF} of Met was assumed to be not significant (Lapierre et al., 2011). The data were collated in an Excel version 2603 workbook (Microsoft Corp.), which included a total of 59 treatment means from 19 studies published from 1989 onward. A summary of publications, identification and coding of treatments is reported in Table 1. Net fluxes reported as grams/12 h (Tagari et al., 2004, 2008) and mmol/h (all other studies) were converted into grams/day using the molecular weight of each EAA. In Tagari et al. (2004, 2008), only net fluxes of free AA were included in the dataset; net fluxes of peptide-bound AA were disregarded because net portal absorption of the sum of free plus peptide-bound

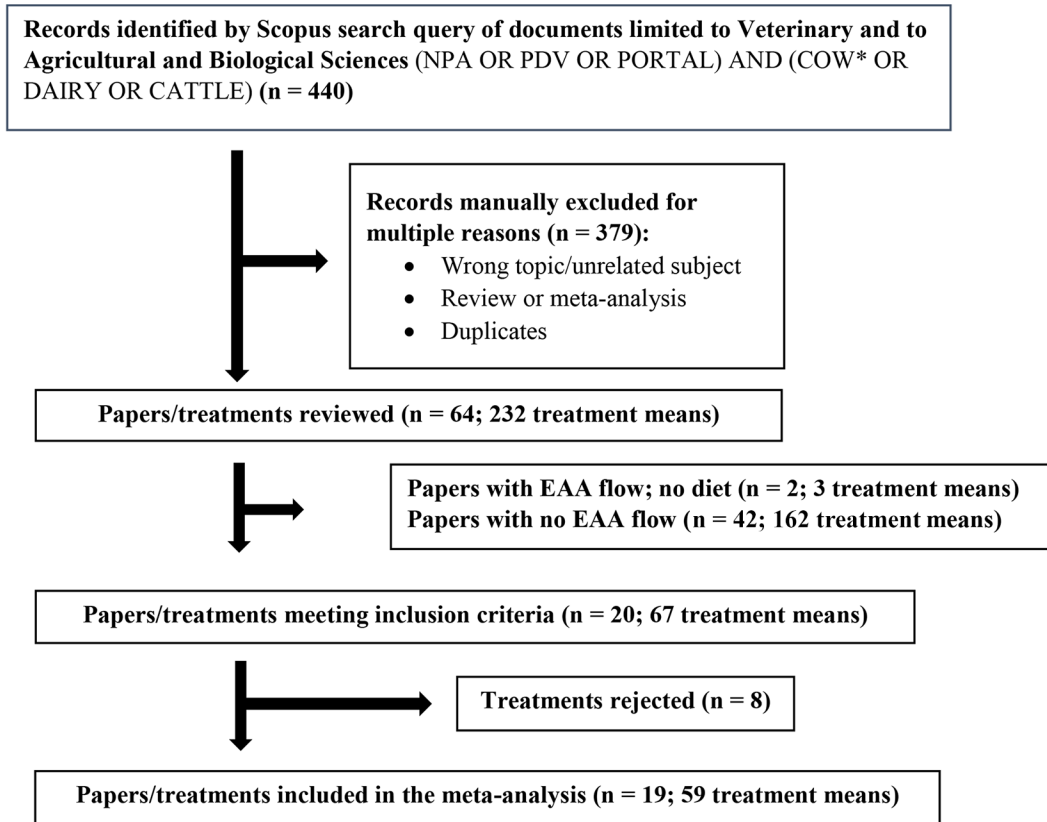


Figure 1. The PRISMA flow diagram of the systematic review from initial search and screening to final selection of publications to be included in the meta-analysis.

EAA absorption was greater than the predicted flow of EAA digested, which biologically cannot occur if model predictions are correct. The semi-essential AA, Arg, was not evaluated in the current study because EffU of Arg was not estimated in NASEM.

Diet Characterization and Infusion

Diet. The nutrient composition of each diet and estimates of mEAA were calculated using NASEM (2021). Feeds matching their description in the study were selected from the NASEM feed library. When reported in studies, the nutrient composition (including AA) of feed ingredients was extracted to create the diet.

Infusion. The dataset included 5 studies with nutrients infused into the abomasum: Doepel et al. (2007; Gln), Lapierre et al. (2009; AA mixture), Larsen et al. (2015; AA mixture), Omphalius et al. (2020; AA mixture and glucose), and Lapierre et al. (2021; AA mixture). Complete details on infusions are reported in Martineau et al. (2025). Infused AA were expressed as grams of free AA and added to NASEM mEAA estimates. The digestible energy supplied by infused nutrients was calculated assuming respectively 2.75 and 1.58 Mcal of NEL per

kilogram of DM for glucose and AA, and 1.73 Mcal of digestible energy per Mcal of NEL (Martineau et al., 2025).

Standardization of the Variables

Different analytical methodologies were used to determine net flux data in the studies included in this meta-analysis. Therefore, some variables were standardized (std) across studies.

Correction of Splanchnic Net Flux Data if *p*-Aminohippuric Acid Was Not Deacetylated. All studies in the dataset used the downstream dilution of *p*-aminohippuric acid (pAH) to estimate the plasma flow across the PDV and splanchnic tissues. Deacetylation of pAH in plasma samples was not reported in Blouin et al. (2002), Lapierre et al. (2004, 2009), Raggio et al. (2004), Tagari et al. (2004, 2008), Berthiaume et al. (2006), and Doepel et al. (2007, 2009). In sheep, Katz and Bergman (1969) estimated that approximately 13% of infused pAH was acetylated when passing through the liver. If pAH is not deacetylated in plasma samples, this would underestimate pAH concentrations, and consequently overestimate splanchnic plasma flow, notably the contribution from

the hepatic arterial plasma flow. In cattle, deacetylation of pAH decreased the estimation of splanchnic plasma flow by 9% (Rodríguez-López et al., 2014). Accordingly, reported TSP_{NF} were multiplied by 0.91 if pAH had not been reported to be deacetylated. In addition, hepatic net fluxes (HEP_{NF}) were recalculated in all studies as the difference between TSP_{NF} (as reported or multiplied by 0.91 if needed) and reported PDV_{NF} .

Essential AA in Milk Proteins. The amount of EAA in MPY was estimated by multiplying reported MPY and the EAA profile in milk true protein (TP; see Table 6-2 in NASEM, 2021). The proportion of blood-borne proteins secreted in milk (e.g., albumin + immunoglobulins) was assumed to be 3.23 g/100 g TP based on data reported in Lapierre et al. (2020). Therefore, the amount of EAA in $MilkAA_{MG}$ (i.e., MPY excluding blood-borne proteins) was calculated by multiplying the milk protein secretion synthesized in the mammary gland and the EAA profile from milk protein fractions, excluding blood-borne proteins based on Lapierre et al. (2020): 2.92, 6.25, 10.53, 8.83, 3.07, 5.23, 4.56, 1.60, 5.86, and 6.91 g/100 g for His, Ile, Leu, Lys, Met, Phe, Thr, Trp, Tyr, and Val, respectively.

Standardized Mammary Net Flux of EAA. A standardization of reported mammary gland net fluxes (MG_{NF}) of EAA was done across studies. The mammary plasma flow (Mpf; L/d) was first standardized based on common Phe and Tyr concentrations in $MilkAA_{MG}$ across studies with the following 2 equations, assuming a U:O ratio of 1. The mammary venous-arterial difference (VAdiff; g/L) of Phe+Tyr in each treatment was calculated as follows:

$$\text{mammary VAdiff of Phe+Tyr} = \left(\frac{\text{reported } MG_{NF} \text{ of Phe+Tyr}}{\text{reported Mpf}} \right) \quad [1]$$

Then, the std Mpf (L/d) in each study was computed as follows:

$$\text{std Mpf} = \left[\frac{\text{reported MPY} \times (1 - 0.0323) \times (5.23 + 5.86)/100}{\text{mammary VAdiff of Phe+Tyr}} \right] \quad [2]$$

where 0.0323 is the proportion of blood-borne proteins in MPY and 5.23 and 5.86 are, respectively, the concentrations (g/100 g TP) of Phe and Tyr in $MilkAA_{MG}$ (Lapierre et al., 2020). Then, the std MG_{NF} (g/d) of each EAA in each treatment was calculated as follows:

$$\begin{aligned} \text{std } MG_{NF} &= \text{std Mpf of each treatment} \\ &\times \text{mammary VAdiff of each EAA.} \end{aligned} \quad [3]$$

Preliminary Calculations

Ratio U:O. The U:O represents the ratio of mammary uptake to $MilkAA_{MG}$ of each EAA. The U:O was standardized as follows:

$$U:O = \frac{-(\text{std } MG_{NF})}{MilkAA_{MG}} \quad [4]$$

For U:O greater than and significantly different from unity, the excess of EAA extracted by the mammary gland relative to $MilkAA_{MG}$ was assumed to be catabolized locally within the tissue.

Ratio $MilkAA_{MG}:TSP_{NF}$. The ratio of $MilkAA_{MG}:TSP_{NF}$ was calculated for each EAA to allow visual data exploration and the selection of treatment means used in the statistical analyses. Visual appraisal and basic statistics led to the stratification of treatment means into 2 stages of lactation: cows in early lactation (DIM 1–30) and in later lactation (DIM 31–305).

Endogenous Supply of EAA. The exploration of $MilkAA_{MG}:TSP_{NF}$ ratios indicated that they were greater than 1 for group 1 AA in early lactation cows. This led us to acknowledge an endogenous supply of EAA (**EndoSupply**) during the first 30 d postpartum only, which would correspond to the uterine involution and the net protein mobilization from muscles. This was based on the assumption that the sum of postliver supply and EndoSupply must cover, at least, mammary uptake plus scurf, plus 3-methyl-His (**3MH**; g/d as calculated from NASEM, 2021) loss in EndoUri for His, or plus EndoUri for group 2 AA. Therefore, the EndoSupply (g/d) was estimated as follows:

$$\text{EndoSupply} = \begin{cases} \text{if DIM} > 30 \text{ or calculations below} < 0 \\ \quad \text{assumed to be 0} \\ \text{else, if DIM} \leq 30 \text{ and His} \\ \quad (-TSP_{NF}) + (U:O \times MilkAA_{MG}) + \text{scurf} + 3MH \\ \text{else, if DIM} \leq 30 \text{ and Met, Phe, or Trp} \\ \quad (-TSP_{NF}) + (U:O \times MilkAA_{MG}) + \text{scurf} \\ \text{else, if DIM} \leq 30 \text{ and the BCAA, Lys, or Thr} \\ \quad (-TSP_{NF}) + (U:O \times MilkAA_{MG}) + \text{EndoUri} + \text{scurf} \end{cases} \quad [5]$$

assuming that urinary 3MH is sourced mainly from muscle tissue degradation. For cows in later lactation, it was assumed that EndoSupply was not present, based on the fact that postliver supplies of His, Phe, and Trp were sufficient to cover $MilkAA_{MG}$. For DIM ≤ 30 , mammary uptake was reported only for Doepel et al.

Table 1. Summary of publications and identification and coding of treatments

Publication	Location	Identification and coding of treatments		
		Publication	Experiment	Treatment
Bach et al. (2000)	MN, USA	1	1.0	Low vs. high CP; low-quality AA profile
Berthiaume et al. (2006)	QC, Canada	2	2.0	Control treatment
Blouin et al. (2002)	QC, Canada	3	3.0	Low MP vs. high MP
Cohick (1989)	NY, USA	4	4.0	1 d pre-bST vs. 7 and 10 d during bST treatment (172 DIM)
Dalbach et al. (2011)	Denmark	5	5.1	Control; 4 vs. 15 vs. 29 DIM
			5.2	HMBi; 4 vs. 15 vs. 29 DIM
Doepel et al. (2007)	QC, Canada	6	6.0	Control vs. abomasal infusion of Gln (300 g/d); 23 DIM
Doepel et al. (2009)	QC, Canada	7	7.0	Postcalving (36 DIM)
Galindo et al. (2024)	QC, Canada	8	8.0	(Soybean meal + corn silage) vs. (canola meal + corn silage) vs. (canola meal + grass silage)
Lapierre et al. (2004)	QC, Canada	9	9.0	Control vs. urea (15 g/kg of DMI)
Lapierre et al. (2009)	QC, Canada	10	10.0	Abomasal infusion of AA mixture (560 g/d); 0 vs. 50.3 g Lys/d
Lapierre et al. (2021)	QC, Canada	11	11.0	Abomasal infusion of AA mixture (595 g/d); 0 vs. 7.6 vs. 15.2 vs. 22.8 g His/d
Larsen and Kristensen (2009)	Denmark	12	12.0	Control; 4 vs. 15 vs. 29 DIM
Larsen and Kristensen (2012)	Denmark	13	13.1	Alfalfa haylage and glucogenic diet; 4 vs. 15 vs. 29 DIM
			13.2	Glucogenic diet; 4 vs. 15 vs. 29 DIM
			13.3	Ketogenic diet; 4 vs. 15 vs. 29 DIM
Larsen et al. (2015)	QC, Canada	14	14.1	5 DIM; control vs. abomasal infusion of AA-CN mixture
			14.2	15 DIM; control vs. abomasal infusion of AA-CN mixture
			14.3	29 DIM; control vs. abomasal infusion of AA-CN mixture
Omphalius et al. (2020)	QC, Canada	15	15.1	Control; 0 vs. abomasal infusion of AA-CN mixture (695 g/d)
			15.2	Abomasal infusion of glucose (1.454 kg/d); 0 vs. AA-CN mixture (695 g/d)
Raggio et al. (2004)	QC, Canada	16	16.0	Low MP vs. medium MP vs. high MP
Roy (2017)	QC, Canada	17	17.0	Canola meal vs. soybean meal vs. (canola meal + urea)
Tagari et al. (2004)	AZ, USA	18	18.0	Corn diet; steam-rolled vs. steam-flaked
Tagari et al. (2008)	AZ, USA	19	19.0	Sorghum diet; dry-rolled vs. steam-flaked

(2007) and Larsen et al. (2015) and therefore the U:O of EAA could only be computed using Equation [4] for these 2 studies. To estimate EndoSupply in studies with DIM ≤ 30 and missing U:O, the average U:O of EAA from Doepel et al. (2007) and Larsen et al. (2015) were used in Equation [5].

Potential Catabolism in Different Tissues. The calculation of Measured_{EffU} was based on the assumption that all the EAA removed by a tissue, on a net basis, and not used to support proteins secreted from the animal would be catabolized. Based on the anatomical placement of the blood sampling in transorgan studies, 4 compartments could be analyzed for potential catabolism: the PDV, the liver and the postsplanchnic tissues, the latter being further separated into PERI and the mammary gland. To increase the robustness of this approach, allocation of a catabolic function for individual EAA within a tissue required that the mean net values were significantly different from 0.

Catabolism of EAA in PDV. The PDV include the gastric compartments, the intestines, cecum, pancreas, spleen, plus the mesenteric and omental fat. Martineau et al. (2025) reported that the amount of EAA in metabolic fecal protein (MFP) could be removed from mEAA to yield an estimate of PDV_{NF}. This concept was used as a proxy of the potential catabolism in the PDV (PDV_{CATA}), in grams/day, calculated as follows:

$$PDV_{CATA} = \begin{cases} \text{if Lys} \\ (mEAA - 11.6 - MFP) - PDV_{NF} \\ \text{else, if other EAA} \\ (mEAA - MFP) - PDV_{NF} \end{cases}, [6]$$

where 11.6 g/d accounts for the overprediction of the metabolizable supply of Lys by the NASEM model (Martineau et al., 2025). This bias overpredicting Lys supply needs to be corrected to adequately determine Lys fate across the PDV.

Catabolism of EAA in the Liver. The potential catabolism by the liver (HEP_{CATA}) was calculated for all EAA even if, in principle, the catabolism would apply only for group 1 AA (Lapierre et al., 2012). It accounted for the amounts of EAA in blood-borne proteins (MilkAA_{BB}; e.g., albumin + immunoglobulins) computed as the difference between EAA in total milk protein secretion (MilkAA_{TOT}) and MilkAA_{MG}. The potential HEP_{CATA} (g/d) was calculated as follows:

$$HEP_{CATA} = \begin{cases} \text{if His, Met, Phe, or Trp} \\ (-HEP_{NF}) - (MilkAA_{BB} + EndoUri) \\ \text{else, if the BCAA, Lys, or Thr} \\ (-HEP_{NF}) - MilkAA_{BB} \end{cases}, [7]$$

where HEP_{NF} were multiplied by -1 because net fluxes are mostly negative, being calculated with VADiff.

It was assumed that part of EAA removal by the liver is not exclusively attributable to catabolism because blood-borne proteins, although synthesized by the liver, would not contribute to potential catabolism. In addition, His, Met, Phe, and Trp in the EndoUri fraction are subtracted from positive HEP_{NF} in Equation [7] assuming that group 1 AA are mainly, if not solely, removed by the liver, and acknowledging that the efficiency for EndoUri is 100% and EndoUri is subtracted from mEAA supply to compute the $NASEM_{EffU}$ (see the denominator in Equation [12]). In contrast, for the other EAA, it was assumed that the removal of AA to support EndoUri would occur mainly from postplanchnic tissues.

Catabolism of EAA in the Postsplanchnic Tissues. The potential net catabolism in postsplanchnic tissues (**PostTSP_{CATA}**) included the EAA oxidized in nonmammary peripheral tissues (**PERI_{CATA}**; e.g., muscles and skin) and mammary gland (**MG_{CATA}**). The potential **PostTSP_{CATA}** (g/d) was calculated as follows:

$$PostTSP_{CATA} = \begin{cases} \text{if His} \\ TSP_{NF} - (MilkAA_{MG} + scurf + 3MH) \\ \text{else, if Met, Phe, or Trp} \\ TSP_{NF} - (MilkAA_{MG} + scurf) \\ \text{else, if the BCAA, Lys, or Thr} \\ TSP_{NF} - (MilkAA_{MG} + scurf + EndoUri) \end{cases} \quad [8]$$

Catabolism of EAA in the Nonmammary Peripheral Tissues. The potential **PERI_{CATA}** (g/d) was calculated as follows:

$$PERI_{CATA} = \begin{cases} \text{if His} \\ TSP_{NF} - [-(std\ MG_{NF}) + scurf + 3MH] \\ \text{else, if Met, Phe, or Trp} \\ TSP_{NF} - [-(std\ MG_{NF}) + scurf] \\ \text{else, if the BCAA, Lys, or Thr} \\ TSP_{NF} - [-(std\ MG_{NF}) + EndoUri + scurf] \end{cases} \quad [9]$$

In studies not reporting MG_{NF} data, a proxy of $std\ MG_{NF}$ was used in Equation [9]:

$$\text{proxy of } std\ MG_{NF} = -(\text{average } U:O_{\text{each lactation stage}} \times MilkAA_{MG}), \quad [10]$$

where average U:O of each EAA in each lactation stage was derived from Equation [4] based on the selected dataset.

Catabolism of EAA in the Mammary Gland. The potential **MG_{CATA}** (g/d) was calculated as follows:

$$MG_{CATA} = [-(std\ MG_{NF}) - MilkAA_{MG}], \quad [11]$$

where $std\ MG_{NF}$ was multiplied by -1 because values were negative. In studies not reporting MG_{NF} data, MG_{CATA} was calculated using a proxy of $std\ MG_{NF}$ (see Equation [10]).

Predicted Efficiency of Utilization of EAA by NASEM

In the calculation of $NASEM_{EffU}$, AA used for protein secretion and accretion (i.e., MPY, MFP, scurf, growth, and gestation) are scaled to the supply of EAA minus the EndoUri losses, assigned to an efficiency of 100%. In the current study, EAA to support protein accretion for growth or gestation were not taken into account because of insufficient details in the studies used. Therefore, $NASEM_{EffU}$ (%) was calculated by scaling EAA in scurf, MFP, and MPY as a function of total EAA supply discounting AA used for EndoUri (all in g/d) as follows:

$$NASEM_{EffU} = \begin{cases} \text{if His} \\ \left(\frac{scurf + MFP + MilkAA_{TOT}}{mEAA + EndoSupply - EndoUri - 3MH} \right) \times 100 \\ \text{else, if Lys} \\ \left(\frac{scurf + MFP + MilkAA_{TOT}}{mEAA - 11.6 + EndoSupply - EndoUri} \right) \times 100 \\ \text{else, if other EAA} \\ \left(\frac{scurf + MFP + MilkAA_{TOT}}{mEAA + EndoSupply - EndoUri} \right) \times 100 \end{cases} \quad [12]$$

where EndoSupply was taken into account for cows in early lactation only and 11.6 g/d accounted for the overprediction of metabolizable Lys by the NASEM model (Martineau et al., 2024, 2025). This bias overpredicting Lys supply needed to be corrected to compute $Measure_{dEffU}$ and $NASEM_{EffU}$ of Lys on the same basis.

Measured Efficiency of Each EAA

The $Measured_{EffU}$ of each EAA was calculated according to catabolism characteristics (i.e., tissues showing a potential catabolism significantly different from 0). Because NASEM assumes an $EffU$ of 100% for EndoUri, this estimated output was subtracted from the EAA net removal to account only for secretions assumed to have a variable $EffU$, as adopted in NASEM (2021). As considered for the calculation of $NASEM_{EffU}$ (Equation [12]), the contribution from EndoSupply was taken into account only for cows in early lactation in the equations in the following sections.

Efficiency of Utilization of Group 1 AA. Based on the significance of the potential catabolism occurring only in the liver, the $\text{Measured}_{\text{EffU}}$ (%) of group 1 AA was calculated by scaling HEP_{CATA} as a function of the total EAA supply minus AA used for EndoUri (all in g/d) as follows:

$$\begin{aligned} & \text{Measured}_{\text{EffU}} \text{ of group 1 AA} \\ &= \begin{cases} \text{if His} \\ \left[1 - \left(\frac{\text{HEP}_{\text{CATA}}}{\text{mEAA} + \text{EndoSupply} - \text{EndoUri} - 3\text{MH}} \right) \right] \times 100 \\ \text{else, if Met, Phe, or Trp} \\ \left[1 - \left(\frac{\text{HEP}_{\text{CATA}}}{\text{mEAA} + \text{EndoSupply} - \text{EndoUri}} \right) \right] \times 100 \end{cases} \end{aligned} \quad [13]$$

Efficiency of Utilization of the BCAA. With potential catabolism occurring in the PDV and mammary gland for the 2 stages of lactation but also in PERI for cows in later lactation, $\text{Measured}_{\text{EffU}}$ of the BCAA were calculated on a stage-of-lactation basis. Indeed, the $\text{PostTSP}_{\text{CATA}}$ for cows with $\text{DIM} \leq 30$ includes a contribution from EndoSupply hiding MG_{CATA} . Therefore, the $\text{Measured}_{\text{EffU}}$ (%) of the BCAA was calculated by scaling the sum of PDV_{CATA} and either MG_{CATA} or $\text{PostTSP}_{\text{CATA}}$ as a function of the total EAA supply minus AA used for EndoUri (all in g/d) as follows:

$$\begin{aligned} & \text{Measured}_{\text{EffU}} \text{ of the BCAA} \\ &= \begin{cases} \text{if } \text{DIM} \leq 30 \\ \left[1 - \left(\frac{\text{PDV}_{\text{CATA}} + \text{MG}_{\text{CATA}}}{\text{mEAA} + \text{EndoSupply} - \text{EndoUri}} \right) \right] \times 100 \\ \text{else, if } \text{DIM} > 30 \\ \left[1 - \left(\frac{\text{PDV}_{\text{CATA}} + \text{PostTSP}_{\text{CATA}}}{\text{mEAA} - \text{EndoUri}} \right) \right] \times 100 \end{cases} \end{aligned} \quad [14]$$

Efficiency of Utilization of Lys. The $\text{Measured}_{\text{EffU}}$ of Lys was calculated acknowledging the potential catabolism of Lys across the liver and in the mammary gland for the 2 stages of lactation but also in PERI for cows in later lactation, similar to the BCAA. Therefore, the $\text{Measured}_{\text{EffU}}$ (%) of the Lys was calculated by scaling the sum of HEP_{CATA} and either MG_{CATA} or $\text{PostTSP}_{\text{CATA}}$ as a function of the total Lys supply minus Lys used for EndoUri (all in g/d) as follows:

$$\begin{aligned} & \text{Measured}_{\text{EffU}} \text{ of Lys} \\ &= \begin{cases} \text{if } \text{DIM} \leq 30 \\ \left[1 - \left(\frac{\text{HEP}_{\text{CATA}} + \text{MG}_{\text{CATA}}}{\text{mEAA} + \text{EndoSupply} - \text{EndoUri} - 11.6} \right) \right] \times 100 \\ \text{else, if } \text{DIM} > 30 \\ \left[1 - \left(\frac{\text{HEP}_{\text{CATA}} + \text{PostTSP}_{\text{CATA}}}{\text{mEAA} - \text{EndoUri} - 11.6} \right) \right] \times 100 \end{cases} \end{aligned} \quad [15]$$

where 11.6 g/d accounts for the overprediction of metabolizable supply Lys by the NASEM model (Martineau et al., 2025).

Efficiency of Utilization of Thr. The $\text{Measured}_{\text{EffU}}$ of Thr was calculated only for cows in later lactation, taking into account the potential catabolism of Thr across the PDV, the liver, and the mammary gland. The $\text{Measured}_{\text{EffU}}$ of Thr could not be calculated for cows in early lactation because, mathematically, $\text{NASEM}_{\text{EffU}}$ of Thr was equivalent to its $\text{Measured}_{\text{EffU}}$. Therefore, the $\text{Measured}_{\text{EffU}}$ (%) of Thr was calculated only for cows later in lactation, scaling the sum of PDV_{CATA} , HEP_{CATA} , and MG_{CATA} as a function of the total Thr supply minus EndoUri (all in g/d) as follows:

$$\begin{aligned} & \text{Measured}_{\text{EffU}} \text{ of Thr} \\ &= \left[1 - \left(\frac{\text{PDV}_{\text{CATA}} + \text{HEP}_{\text{CATA}} + \text{MG}_{\text{CATA}}}{\text{mEAA} + \text{EndoSupply} - \text{EndoUri}} \right) \right] \times 100. \end{aligned} \quad [16]$$

Adjusted $\text{Measured}_{\text{EffU}}$ of Each EAA

Given the scant data available, the inherent variability associated with reported net flux data, plus the differences in experimental design between studies, $\text{Measured}_{\text{EffU}}$ values were adjusted for the random effects of study, as reported in St-Pierre (2001), to reflect the ability of NASEM to predict changes in $\text{Measured}_{\text{EffU}}$ on a within-study basis. Briefly, (1) $\text{Measured}_{\text{EffU}}$ were regressed on $\text{NASEM}_{\text{EffU}}$ in a model adjusted for the random study effects with observations weighted by the std SEM using the lmer function in R (version 4.5.2), (2) fitted $Y = \text{intercept} + (\text{slope} \times \text{NASEM}_{\text{EffU}})$, and (3) conditional residuals were extracted from the regression model and added to fitted Y values. More details on this procedure are reported in Martineau et al. (2023, 2024). As observed in the latter meta-analyses, this adjustment had a small effect on averages of adjusted values but reduced SD.

Data Weighting

In Cohick (1989), data were averaged from 3 cows and the std SEM was 1.0 by default. All other studies reported SEM in fixed-effects models (7 studies and 15 treatment means) or mixed-effects models in remaining studies. The std SEM accounted for the type of statistical model used in studies as reported in Martineau et al. (2023, 2024).

Evaluation of Predictions

Selection of Treatment Means. For VAdiff measurements, analytical precision is required, especially regarding the CV, which need to be lower than the measured VAdiff for each AA. Measuring AA concentration using isotope-dilution GC/MS allowed decreasing the CV below 1% (Calder et al., 1999). Threonine presents an additional challenge because in conventional approaches relying on chromatography, Thr often does not achieve baseline resolution with other AA such as Ser (Lautenschläger et al., 2017) or Ala and Arg (White et al., 1986). For this reason, the data selected were from treatments where the isotope-dilution technique had been used and 8 Thr net fluxes from 4 studies (Cohick, 1989; Blouin et al., 2002; Tagari et al., 2004, 2008) were not included in the analysis. In addition, 2 His net fluxes (Larsen and Kristensen, 2009, control, 29 DIM and Larsen and Kristensen, 2012, ALF-GLCG diet, 15 DIM) and 1 Trp net flux (Larsen and Kristensen, 2012, KETO diet, 15 DIM) were not included in the analysis because HEP_{NF} was positive. The remaining data were used as the initial dataset.

The calculation of $Measured_{EffU}$ of EAA cumulates errors associated with the measurement of splanchnic and mammary gland net fluxes and the estimation of endogenous supply in early lactating cows. Therefore, a statistical approach was used to identify outliers and select objectively the treatment means to be used in the statistical analyses. The interquartile range (IQR) method is a measure of statistical dispersion defined as the difference between the third and first quantiles (Upton and Cook, 1996). Values falling outside first quantile minus ($1.5 \times IQR$) or third quantile plus ($1.5 \times IQR$) were vetted for data entry errors on our part; if not, they were considered outliers and removed from the initial dataset. The IQR method was used on a stage-of-lactation basis for 2 variables: (1) the ratio $MilkAA_{MG}:TSP_{NF}$, and (2) the potential catabolism of each EAA used to compute its corresponding $Measured_{EffU}$. The ratio $MilkAA_{MG}:TSP_{NF}$ was assumed to be the best variable available for the selection of treatment means because it integrates (1) $MilkAA_{MG}$, a robust measure independent of net flux measurement, and (2) TSP_{NF} , the sum of PDV_{NF} and HEP_{NF} . In early lactation, high

$MilkAA_{MG}:TSP_{NF}$ ratios can be associated with small TSP_{NF} values, indicating contribution from endogenous supply of EAA to support $MilkAA_{MG}$; nonetheless, these $MilkAA_{MG}:TSP_{NF}$ ratios should be biologically plausible. Treatments with skewed estimates of potential catabolism of EAA were not included in the selected dataset because this variable accumulated the errors from net fluxes across the different organs as shown in Equations [6], [7], and [8], or Equation [11]. The list of treatments means removed from the initial dataset is reported Supplemental Table S1 (see Notes), with the reason for their removal. The remaining treatment means were used as the selected dataset for the statistical analyses.

Descriptive and Fit Statistics. The concordance correlation coefficient (CCC; Lin, 1989) was extracted from the `epi.ccc` function of the `epiR` package in R version 2.0.80. The mean square prediction error (MSPE) was calculated as follows:

$$MSPE = \frac{1}{n} \times \sum_{i=1}^n (O_i - P_i)^2, \quad [17]$$

where n = number of treatment means, O_i = i th adjusted $Measured_{EffU}$ value, and P_i = i th corresponding $NASEM_{EffU}$ value. The square root of MSPE (**RMSPE**) and the relative prediction error (**RPE**; RMSPE as a percentage of the average of observed values [**obs.mean**]) were used to assess the goodness-of-fit of $NASEM_{EffU}$. The MSPE was decomposed into error in central tendency (**ECT**), error due to deviation of the regression slope from unity (**ER**), and error due to disturbances (not reported; Theil, 1966; Bibby and Toutenburg, 1978). The ratio of RMSPE to SD of measured data (**RSR**) was also computed, and the smaller the better for the model performance (Moriassi et al., 2007).

Mean and Linear Biases. The mean bias and its significance were, respectively, the mean and the Student's t -test P -value of residual errors (i.e., $Measured_{EffU} - NASEM_{EffU}$). The degree of agreement between both $EffU$ was assessed by visualizing plots created by the `ggplot2` package (version 4.0.0 in R; Wickham, 2009).

The other bias, the linear bias, is related to the slope between $Measured_{EffU}$ and $NASEM_{EffU}$. This bias is always positive and independent of the mean bias. More details on its calculation are reported in Martineau et al. (2023, 2024). To avoid type I errors and given the variability of transorgan net flux data, mean and linear biases were considered within acceptable limits below a threshold of $10.0\%_{obs.mean}$, even if significant. At $\geq 10.0\%_{obs.mean}$, biases were deemed compromised both in statistical and physiological terms.

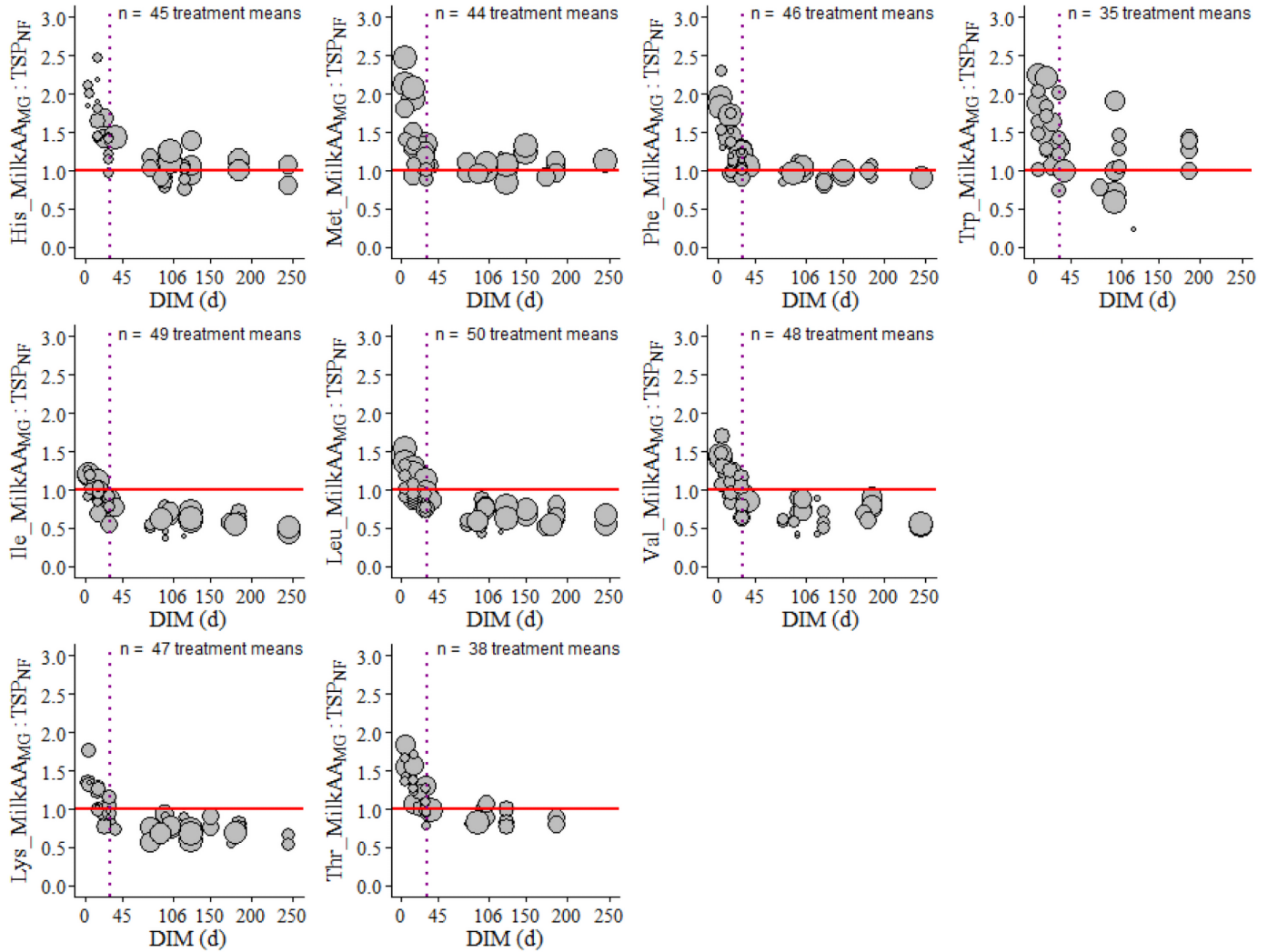


Figure 2. Relationships between the ratio $\text{MilkAA}_{\text{MG}}:\text{TSP}_{\text{NF}}$ of EAA versus DIM for the studies included in the selected dataset (refer to Table 2). Symbols are sized according to the standardized SEM. The vertical dotted line represents d 31.

Statistical Model

The hierarchical structure of data (i.e., treatment/experiment/publication) was accounted for in statistical models. For example, a single publication could contain one or more experiments (e.g., Larsen and Kristensen, 2012). The data were fitted to multilevel mixed-effect meta-regression models using the `rma.mv` of the `metafor` package and the `robust` function in R (version 4.8-0; REML method; Viechtbauer, 2010, 2018). Unbiased estimates of fixed effects and valid estimates of *P*-values are obtained using the `robust` function (Viechtbauer, 2021). The linear model can be written statistically as

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + u_i + w_{ij} + e_{ij},$$

$$u_i \sim N\left(0, \tau_{\text{experiment}}^2\right), w_{ij} \sim N\left(0, \tau_{\text{publication}}^2\right), \quad [18]$$

where y_{ij} is the observed value for the i th experiment ($i = 1, \dots, I_j$) clustered in the j th publication ($j = 1, \dots, J_k$); β_0, β_1 is the fixed effect parameters corresponding to the intercept and x_1 (i.e., the linear term); u_i and w_{ij} are, respectively, the random effects for the residual random heterogeneity at the experiment and publication levels; and e_{ij} is the sampling error. Random effects of experiment and publication are assumed to be independently, identically, and normally distributed with a mean of zero and a variance of $\tau_{\text{experiment}}^2$ and $\tau_{\text{publication}}^2$, respectively. The sampling errors are assumed to be independently and normally distributed with a mean of zero and a variance of v_{ij} . Therefore, the v_{ij} are the (approximately) known and heteroscedastic sampling variances of the dependent variable (W. Viechtbauer, Department of Psychiatry and Neuropsychology, Maastricht University, Maastricht, the

Table 2. Ratios of MilkAA_{MG}:TSP_{NF} of EAA for the 2 stages of lactation in the initial and the selected datasets¹

Variable	Cows with DIM 1–30						Cows with DIM 31–305					
	n	Mean	Median	SD	Min.	Max.	n	Mean	Median	SD	Min.	Max.
Initial dataset												
His	24	2.17 ^a	1.69	2.166	−2.11	10.38	29	1.22 ^b	1.08	0.671	0.77	4.41
Ile	26	1.03 ^{A,‡}	0.93	0.369	0.56	2.19	27	0.67 ^B	0.62	0.262	0.38	1.71
Leu	26	1.11 ^{A,‡}	0.95	0.455	0.71	2.57	29	0.72 ^B	0.68	0.226	0.42	1.42
Lys	26	1.26 ^A	1.11	0.555	0.61	3.09	29	0.80 ^B	0.74	0.301	0.54	2.17
Met	26	3.52 [‡]	1.34	13.852	−14.20	69.69	29	1.15 [‡]	1.07	0.531	0.49	3.15
Phe	26	2.17	1.28	3.133	0.91	16.39	29	1.00 [‡]	1.00	0.281	0.48	1.83
Thr	26	6.33 [‡]	1.27	25.163	0.69	129.66	19	1.10 [‡]	0.87	0.703	0.51	3.80
Trp	22	1.77	1.55	0.844	0.75	4.63	21	2.02 [‡]	1.01	4.853	−1.62	22.92
Val	26	1.21 ^A	1.09	0.501	0.64	2.94	27	0.76 ^B	0.69	0.333	0.39	1.95
Selected dataset												
His	19	1.64 ^A	1.64	0.387	0.96	2.47	26	1.05 ^{B,‡}	1.04	0.170	0.77	1.42
Ile	24	0.94 ^A	0.92	0.176	0.56	1.25	25	0.61 ^B	0.61	0.115	0.38	0.80
Leu	24	1.00 ^{A,‡}	0.94	0.223	0.71	1.54	26	0.66 ^B	0.67	0.124	0.42	0.88
Lys	20	1.12 ^A	1.06	0.233	0.77	1.76	27	0.73 ^B	0.74	0.106	0.54	0.94
Met	22	1.41 ^A	1.33	0.421	0.89	2.47	22	1.06 ^B	1.07	0.107	0.84	1.33
Phe	23	1.36 ^A	1.25	0.356	0.91	2.31	23	0.98 ^{B,‡}	1.00	0.084	0.81	1.13
Thr	21	1.28 ^A	1.26	0.275	0.79	1.84	17	0.86 ^B	0.86	0.095	0.77	1.06
Trp	20	1.54 ^A	1.48	0.409	0.75	2.25	15	1.08 ^{B,‡}	1.01	0.411	0.23	1.92
Val	23	1.08 ^{A,‡}	1.07	0.273	0.64	1.70	25	0.68 ^B	0.69	0.158	0.39	0.91

^{a,b}Within rows, values with different superscripts differ ($P \leq 0.05$).

^{A,B}Within rows, values with different superscripts differ ($P \leq 0.01$).

¹n = number of treatment means before (initial dataset) and after the removal of outliers (selected dataset); Min. = minimum; Max. = maximum.

[‡]Mean is not different ($P > 0.10$) from 1.

Netherlands, personal communication). One- and 2-sample *t*-tests were performed using the *t.test* function in R. Statistical significance was declared at P -value ≤ 0.05 and a tendency at $0.05 < P$ -value ≤ 0.10 .

RESULTS

Features of the Dataset

The variables reported in all treatment means in the dataset included DIM, DMI, intake and concentration of CP, yields of milk, and concentration and yield of milk protein. Milk protein concentration and yield were reported on a CP basis in 5 studies (10 treatment means) and converted into a TP basis by multiplying by 0.951 (DePeters and Cant, 1992). During plasma sampling, DIM averaged 79 d and ranged from 4 to 245 d. Data for cows in early lactation were collected from Doepel et al. (2007), Larsen and Kristensen (2009), Dalbach et al. (2011), Larsen and Kristensen (2012), and Larsen et al. (2015). In Doepel et al. (2007), plasma was sampled at 23 DIM whereas it was collected sequentially at 4 or 5, 15, and 29 DIM in the 4 Danish studies (Table 1).

The Ratio MilkAA_{MG}:TSP_{NF}. An initial graphical appraisal of the ratios MilkAA_{MG}:TSP_{NF} in the initial dataset indicated that DIM created 2 different populations of treatment means (Figure 2). For cows with DIM ≤ 30 (cows in early lactation from here on), the ratios MilkAA_{MG}:TSP_{NF} of group 1 AA were greater ($P < 0.001$) than 1

with averages ranging from 1.36 to 1.64 (selected dataset in Table 2). In contrast, for cows with DIM > 30 (cows in later lactation from here on), the averages of the ratios MilkAA_{MG}:TSP_{NF} of group 1 AA ranged from 0.98 to 1.08 and were not different ($P \geq 0.18$) from 1 for His, Phe, and Trp, although being higher ($P = 0.01$) than 1 for Met.

For cows in early lactation, the averages of the ratios MilkAA_{MG}:TSP_{NF} of the BCAA, Lys, and Thr ranged from 0.94 to 1.28, and were not different from 1 for the BCAA (except for Ile at $P = 0.09$), but were greater ($P \leq 0.03$) than 1 for Lys and Thr (Table 2). In contrast, for cows in later lactation, the averages of the ratios MilkAA_{MG}:TSP_{NF} of BCAA, Lys, and Thr ranged from 0.61 to 0.86 and were all different ($P < 0.001$) from 1. A ratio MilkAA_{MG}:TSP_{NF} > 1 , typical of group 1 AA and Thr for cows in early lactation, suggests that endogenous sources supplied the missing EAA needed to support the observed MilkAA_{MG}. Because of the impact of this contribution from endogenous sources on the calculation of EffU, treatment means were stratified by stages of lactation.

The statistical IQR test identified 50 outliers based on the MilkAA_{MG}:TSP_{NF} ratio, 11 outliers based on potential EAA catabolism, and 4 outliers based on both criteria (Supplemental Table S1). A visual appraisal of the ratios MilkAA_{MG}:TSP_{NF} confirmed this identification, with those treatment means clearly being out of the clusters of the treatment means of the selected database.

Reported Milk Data. A summary of animal and dietary characteristics reported in publications is presented in

Table 3. Reported animal and dietary characteristics for the 2 stages of lactation in the initial dataset^{1,2}

Variable	Cows with DIM 1–30						Cows with DIM 31–305					
	n	Mean	Median	SD	Min.	Max.	n	Mean	Median	SD	Min.	Max.
Animal characteristics³												
BW (kg)	26	618 ^b	604	58.0	542	733	28	652 ^a	656	41.9	569	733
DIM (d)	26	17 ^b	15	10.1	4	29	33	126 ^A	120	50.7	36	245
Production												
Milk (kg/d)	26	36.3	35.7	7.49	20.0	51.9	33	32.9	33.9	8.25	15.9	48.6
True protein (g/d)	26	1,186 ^A	1,185	153.8	840	1,502	33	984 ^B	970	208.5	621	1,425
Milk concentration (%)												
True protein ⁴	26	3.34 ^a	3.20	0.494	2.60	4.30	33	3.09 ^b	2.95	0.472	2.35	4.41
Fat	26	4.21 ^A	4.20	0.510	3.40	5.40	24	3.63 ^B	3.66	0.371	2.90	4.25
Lactose	26	4.66	4.66	0.213	4.21	5.00	18	4.57	4.61	0.422	3.75	5.00
Diet characteristics												
Dietary DM (kg/d)	26	17.4 ^B	18.0	2.84	11.9	23.0	33	20.9 ^A	20.1	3.26	16.3	27.0
Infused DM (kg/d)	4	0.47 ^y	0.43	0.258	0.23	0.79	9	0.88 ^x	0.61	0.553	0.56	2.15
Total DMI (kg/d)	26	17.4 ^B	18.0	2.87	11.9	23.2	33	21.1 ^A	20.7	3.16	16.3	27.0
CP (% of DM)	26	16.8 ^A	16.4	1.21	15.5	18.7	33	15.2 ^B	16.1	2.10	12.0	19.1
NDF (% of DM)	26	25.9 ^B	27.3	8.11	15.7	36.1	21	32.6 ^A	31.5	4.30	28.3	40.8
Dietary N (g/d)	26	465 ^z	484	74.3	312	604	33	509 ^x	482	120.5	369	795
Infused N (g/d)	4	68	64	36.0	32	113	9	85	90	33.2	0	111
Total N intake (g/d)	26	475 ^b	487	81.4	312	636	33	532 ^a	488	102.5	369	795
Milk N efficiency ⁵ (%)	26	0.42 ^A	0.40	0.065	0.34	0.57	33	0.31 ^B	0.31	0.052	0.22	0.40

^{a,b}Within rows, values with different superscripts differ ($P \leq 0.05$).

^{x,y}Within rows, values with different superscripts tended ($P \leq 0.10$) to differ.

^{A,B}Within rows, values with different superscripts differ ($P \leq 0.01$).

¹Data as reported in the publications for cows in early (DIM 1–30) and later lactation (DIM 31–305).

²n = number of treatment means; Min. = minimum; Max. = maximum.

³BW = 618 kg and DIM = 150 d in H. Lapierre and D.R. Ouellet: Agriculture and Agri-Food Canada, Sherbrooke, QC, Canada (personal communication). The BW was missing in Tagari et al. (2004, 2008) and Doepel et al. (2009), and assumed to be 630 kg (average BW of cows in the dataset).

⁴If not reported, milk protein was assumed to be CP for publications earlier than 1990 and true protein thereafter. Milk true protein = milk CP $\times$ 0.951 (DePeters and Cant, 1992).

⁵Milk N efficiency (%) = (milk CP yield/6.38)/N intake $\times$ 100.

Table 3. All cows were multiparous Holstein-Friesian (personal communication to Lapierre et al., 2004), except for cows in Tagari et al. (2004), which included 4 primiparous and 2 second-lactation cows, Tagari et al. (2008), which included 6 primiparous and 2 second-lactation cows, and Berthiaume et al. (2006), which included 3 primiparous and 3 multiparous cows. Milk yield averaged 34.4 kg/d (SD = 8.03, range 15.9–51.9) across treatments and did not differ ($P = 0.11$) between stages of lactation. The following production data were on average greater ($P \leq 0.05$) for cows in early versus later lactation: MPY (17%) and milk concentrations of TP (8%) and fat (14%).

Reported Diet Characteristics. The total DMI (diet + infusion) was 18% smaller ($P < 0.001$) for cows in early versus in later lactation (Table 3). Dietary CP concentration was 10% greater ($P < 0.001$) for cows in early versus later lactation. Dietary N intake tended ($P = 0.09$) to be 9% smaller for cows in early versus later lactation. Total N intake (diet + infusion) was 11% smaller ($P = 0.02$) for cows in early versus later lactation. Calculated milk N efficiency relative to N intake was 26% greater ($P < 0.001$) for cows in early versus later lactation, which further supports the stratification

of treatment means by stages of lactation in the current study.

Diets Reconstituted by NASEM. Across diets, concentrations of CP, NFC, and digestible energy in NASEM-reconstituted diets were, on average, 3% to 10% greater ($P \leq 0.03$) for cows in early compared with those in later lactation (Table 4). Total N intake (diet + infusion) reported in publications (507 g/d) and reconstituted diets (517 g/d) were similar ($P = 0.61$), and the latter was 10% ($P = 0.03$) smaller for cows in early versus later lactation. Estimated MP supply averaged 1,939 g/d (SD = 318.1, range 1,164–2,542) across treatments. Despite MP supply being similar ($P = 0.18$) between the 2 stages of lactation, mEAA of Ile, Lys, Thr, Trp, and Val were on average 8% to 12% smaller ($P \leq 0.04$) and mEAA of His and Phe tended ($P \leq 0.07$) to be smaller for cows in early versus later lactation. In contrast, mEAA of Leu and Met were similar ($P \geq 0.17$) between stages of lactation.

Plasma Flow and Net Flux Data. The data on reported net fluxes of EAA (mmol/h) are available in Supplemental Table S2 (see Notes). In other tables, net fluxes of EAA were converted into a grams/day basis. In addition, HEP_{NF} and TSP_{NF} were corrected if pAH had not been deacetylated (see preliminary calculations in Materi-

Table 4. Dietary characteristics computed by NASEM (2021) for the 2 stages of lactation in the initial dataset^{1,2}

Variable	Cows with DIM 1–30						Cows with DIM 31–305					
	n	Mean	Median	SD	Min.	Max.	n	Mean	Median	SD	Min.	Max.
Diet concentration												
CP (% of DM)	26	17.2 ^A	16.5	1.87	14.7	20.6	33	15.4 ^B	16.2	2.42	12.1	19.6
NDF (% of DM)	26	28.0 ^b	27.8	7.72	17.5	38.6	33	32.0 ^a	32.3	5.52	21.5	40.9
NFC (% of DM)	26	45.2 ^a	47.9	7.13	34.8	53.5	33	41.6 ^b	42.3	4.51	33.4	51.2
Digestible energy (Mcal/kg)	26	3.10 ^A	3.14	0.168	2.87	3.35	33	3.00 ^B	3.01	0.093	2.85	3.16
Total N intake (g/d)	26	486 ^b	494	84.7	294	642	33	541 ^a	502	101.6	369	792
Metabolizable supply (g/d)												
MP	26	1,874	1,856	352.1	1,164	2,542	33	1,990	2,053	283.4	1,402	2,480
His	26	43.6 ^y	43.8	9.19	26.1	58.6	33	47.8 ^x	49.4	8.13	32.9	64.7
Ile	26	107.5 ^b	108.2	17.76	70.8	139.9	33	117.0 ^a	118.6	16.08	85.3	147.6
Leu	26	171.8	167.2	38.26	100.6	248.8	33	183.2	186.9	26.15	123.9	221.2
Lys ³	26	122.2 ^B	124.9	22.28	79.5	166.0	33	139.0 ^A	140.5	22.46	100.7	180.0
Met	26	43.5	43.9	8.64	26.8	58.1	33	46.5	48.5	7.62	32.7	55.9
Phe	26	109.4 ^y	105.1	25.07	66.2	165.7	33	121.1 ^x	117.1	22.14	82.2	154.2
Thr	26	96.0 ^b	96.4	16.13	64.0	123.1	33	105.1 ^a	106.4	14.46	78.1	130.9
Trp	26	23.9 ^B	23.9	4.23	15.1	31.7	33	27.0 ^A	28.2	3.59	20.5	32.9
Val	26	111.9 ^B	112.7	20.08	74.1	147.3	33	125.7 ^A	129.0	18.01	90.7	158.4

^{a,b}Within rows, values with different superscripts differ ($P \leq 0.05$).

^{x,y}Within rows, values with different superscripts tended ($P \leq 0.10$) to differ.

^{A,B}Within rows, values with different superscripts differ ($P \leq 0.01$).

¹Variables computed separately for cows in early (DIM 1–30) and later lactation (DIM 31–305).

²n = number of treatment means; Min. = minimum; Max. = maximum.

³Estimated metabolizable Lys supply corrected for the mean bias (–11.6 g/d) observed in Martineau et al. (2024, 2025).

als and Methods). In Table 5, plasma flows across the PDV and the splanchnic tissues were on average 17% to 18% smaller ($P < 0.001$) for cows in early versus later lactation, and the opposite for Mpf and std Mpf. The standardization process did not alter ($P \geq 0.41$) Mpf with numerical increments of 2% and 4% for cows in early and later lactation, respectively.

The data on net fluxes across tissues and in MilkAA_{MG} are reported for the selected dataset in Table 5. On average, the PDV_{NF} of all EAA were positive ($P < 0.001$), indicating net release of the EAA in the portal circulation. The PDV_{NF} of Ile, Lys, Met, Phe, Trp, and Val were 12% to 23% smaller ($P \leq 0.05$) and that of Thr tended ($P = 0.06$) to be 14% smaller for cows in early versus later lactation. In contrast, PDV_{NF} of His and Leu were similar ($P \geq 0.30$) between stages of lactation.

Overall and on average, HEP_{NF} of the BCAA were either not different from 0 ($P \geq 0.26$) or positive (Ile, $P = 0.04$ and Leu, $P = 0.07$, respectively) for cows in later lactation. In contrast, the HEP_{NF} of the other EAA were all negative ($P < 0.001$), indicating net removal by the liver. The HEP_{NF} of His, Lys, Thr, Trp, and Val did not differ ($P \geq 0.22$) between stages of lactation. However, HEP_{NF} of Ile tended ($P = 0.07$) and that of Leu was different ($P = 0.04$) for cows in early versus later lactation in contrast to those of Met and Phe ($P \leq 0.01$) being in the other direction.

On average, the TSP_{NF} were positive ($P < 0.001$), indicating a release of the EAA across the splanchnic tissues.

The TSP_{NF} of the BCAA, His, Lys, and Thr were 13% to 19% smaller ($P \leq 0.03$) for cows in early versus later lactation whereas those of Met, Phe, and Trp did not differ ($P \geq 0.16$) between stages of lactation.

Only 2 studies reported MG_{NF} in early lactating cows: Doepel et al. (2007) and Larsen et al. (2015). On average, std MG_{NF} were all negative, indicating a mammary net uptake which was 20% to 30% greater ($P \leq 0.005$) for cows in early versus later lactation, except for Val which tended ($P = 0.08$) to be 10% greater in the same direction. The EAA secreted in MilkAA_{MG} were all 18% to 21% greater ($P < 0.001$) for cows in early versus later lactation, similar to the 17% difference in MPY between the 2 stages of lactation (Table 3).

Estimates of Endogenous EAA Supply

The amounts of EAA supplied by endogenous sources estimated in Eq. [5] are detailed in Table 6. The EndoSupply of EAA were all greater ($P < 0.001$) than 0. Plasma was sampled sequentially at 4 or 5, 15, and 29 DIM in Dalbach et al. (2011), Larsen and Kristensen (2009, 2012), Larsen et al. (2015). In these 4 studies, the ratio of EndoSupply on total EAA supply (EndoSupply + mEAA) decreased with increments of DIM: it averaged 0.33 (SD = 0.105, range 0.08 to 0.59) at d 4 or 5, 0.18 (SD = 0.111, range 0.00 to 0.40) at d 15, and 0.10 (SD = 0.095, range 0.00 to 0.38) at d 29 (n = 71; data not shown).

Table 5. Plasma flows (L/h) in the initial dataset, and transorgan net fluxes (g/d) for the 2 stages of lactation in the selected dataset^{1,2}

Variable	Cows with DIM 1–30						Cows with DIM 31–305					
	n	Mean	Median	SD	Min.	Max.	n	Mean	Median	SD	Min.	Max.
Plasma flow												
Portal	26	1,237 ^A	1,243	229.3	778	1,660	31	1,492 ^B	1,474	260.5	1,103	2,209
Splanchnic	26	1,409 ^A	1,347	297.6	873	2,019	31	1,728 ^B	1,676	334.9	1,163	2,395
Mpf	8	813 ^A	793	50.6	748	896	27	661 ^B	669	117.1	491	909
Std Mpf ³	8	830 ^A	828	44.1	785	908	27	689 ^B	669	128.7	496	1,076
PDV _{NF}												
His	19	35.4	35.0	10.83	18.2	57.0	26	38.6	39.1	9.38	22.3	52.9
Ile	24	81.4 ^b	78.7	15.80	56.7	125.9	25	92.3 ^a	91.9	19.44	54.5	128.4
Leu	24	131.5 ^y	129.1	31.39	78.7	204.6	26	146.8 ^x	145.0	33.35	81.9	207.8
Lys	20	105.2 ^b	108.8	21.94	70.2	143.9	27	120.3 ^a	120.7	26.63	72.2	175.4
Met	22	38.3 ^b	35.6	10.62	22.2	60.9	22	45.0 ^a	45.7	10.92	24.7	65.9
Phe	23	90.5 ^b	87.2	21.49	55.5	134.8	23	107.2 ^a	106.3	24.80	57.9	152.2
Thr	21	65.5 ^c	65.8	13.67	45.7	91.5	17	76.0 ^x	78.9	18.16	40.9	106.1
Trp	20	22.7 ^b	19.6	8.08	10.8	36.3	15	29.5 ^a	28.9	10.89	12.7	58.2
Val	23	78.4 ^B	73.1	17.78	56.2	132.1	25	95.1 ^A	94.5	25.66	46.4	143.8
HEP _{NF}												
His	19	−13.2	−12.3	7.30	−26.4	−0.7	26	−12.1	−12.4	6.99	−29.2	0.0
Ile	24	−1.3 ^{y‡}	−1.6	11.08	−22.0	27.1	25	4.0 ^x	6.2	9.06	−25.6	15.3
Leu	24	−2.8 ^{b‡}	−1.6	11.75	−22.0	25.2	26	3.7 ^a	4.7	10.07	−21.1	18.0
Lys	20	−9.1	−8.9	7.05	−24.6	3.5	27	−8.1	−8.8	9.05	−27.7	10.5
Met	22	−11.3 ^A	−10.0	5.51	−22.6	−1.8	22	−17.6 ^B	−18.0	6.59	−26.5	−3.6
Phe	23	−43.6 ^A	−44.4	15.37	−75.3	−21.0	23	−56.7 ^B	−51.3	19.38	−93.5	−20.2
Thr	21	−21.5	−22.6	10.47	−42.9	−2.9	17	−25.3	−23.4	11.86	−46.9	−1.1
Trp	20	−9.5	−6.9	6.44	−20.6	−0.5	15	−11.8	−11.9	4.62	−19.4	−3.6
Val	23	0.0 [‡]	0.0	13.24	−28.0	29.5	25	1.9 [‡]	6.7	13.45	−36.6	20.7
TSP _{NF}												
His	19	22.3 ^b	21.2	6.49	13.8	39.5	26	26.5 ^a	27.1	5.84	16.3	36.1
Ile	24	80.0 ^B	80.5	13.76	59.8	105.8	25	96.3 ^A	93.7	20.18	56.2	125.2
Leu	24	128.7 ^B	130.6	29.30	78.7	188.9	26	150.5 ^A	150.5	29.86	92.8	188.9
Lys	20	96.1 ^b	94.7	20.92	59.6	129.8	27	112.2 ^a	113.3	24.97	71.5	162.5
Met	22	27.0	27.9	7.10	14.3	39.4	22	27.4	29.0	6.07	17.9	39.4
Phe	23	46.9	48.0	10.44	27.0	65.0	23	50.4	52.3	9.83	34.6	69.0
Thr	21	44.0 ^b	42.9	10.07	28.6	65.8	17	50.7 ^a	52.6	8.38	35.6	65.5
Trp	20	13.2	12.0	3.84	7.8	22.1	15	17.7	16.7	11.13	8.8	54.6
Val	23	78.5 ^B	74.5	19.99	55.4	136.1	25	97.0 ^A	101.5	23.46	55.1	128.9
MG _{NF}												
His	7	−37.7 ^B	−37.2	2.96	−42.2	−33.7	24	−29.3 ^A	−29.0	6.02	−38.6	−19.2
Ile	8	−96.7 ^B	−96.7	12.92	−113.4	−78.4	21	−77.4 ^A	−75.1	17.41	−114.1	−56.0
Leu	8	−155.0 ^B	−156.9	24.16	−185.8	−127.0	20	−121.3 ^A	−122.7	24.38	−172.8	−84.3
Lys	7	−124.4 ^B	−122.8	13.76	−144.7	−104.8	21	−97.6 ^A	−97.2	31.29	−148.6	−49.9
Met	7	−39.2 ^B	−39.9	7.11	−48.0	−29.9	18	−29.4 ^A	−28.4	6.64	−41.7	−20.2
Phe	7	−71.5 ^B	−75.3	8.72	−81.7	−62.1	22	−54.3 ^A	−53.3	9.54	−76.4	−38.4
Thr	8	−61.7 ^B	−61.5	7.02	−71.5	−51.3	17	−48.2 ^A	−48.1	9.90	−66.6	−33.9
Trp	8	−24.9 ^B	−24.5	4.73	−31.8	−18.6	15	−17.5 ^A	−16.8	4.71	−25.4	−10.4
Val	7	−89.3 ^y	−89.6	7.60	−98.4	−75.6	21	−80.5 ^x	−80.6	18.05	−114.9	−51.1
MilkAA _{MG}												
His	19	34.3 ^A	33.7	3.38	27.4	40.4	30	27.3 ^B	27.4	5.29	17.5	39.0
Ile	24	73.0 ^A	72.0	8.33	58.7	90.8	31	58.0 ^B	58.7	11.39	37.6	83.4
Leu	24	122.9 ^A	121.3	14.03	98.8	153.1	30	98.4 ^B	98.8	19.06	63.3	140.5
Lys	20	103.9 ^A	102.4	9.59	86.4	122.2	31	81.9 ^B	82.9	16.09	53.1	117.8
Met	22	35.4 ^A	35.2	3.80	28.8	42.5	26	29.1 ^B	28.8	5.53	20.3	41.0
Phe	23	60.4 ^A	60.2	6.34	49.1	72.4	27	49.7 ^B	51.3	9.58	31.4	69.8
Thr	21	54.2 ^A	52.6	5.57	44.6	66.3	23	44.9 ^B	44.9	7.59	31.9	60.9
Trp	23	18.5 ^A	18.4	2.48	13.0	23.3	27	14.9 ^B	15.0	3.04	9.6	21.4
Val	23	80.0 ^A	79.6	8.83	64.9	100.4	31	64.1 ^B	64.8	12.59	41.5	92.2

^{a–c} Within rows, values with different superscripts differ ($P \leq 0.05$).^{x,y} Within rows, values with different superscripts tended ($P \leq 0.10$) to differ.^{A,B} Within rows, values with different superscripts differ ($P \leq 0.01$).¹ Data computed for cows in early (DIM 1–30) and later in lactation (DIM 31–305). Net fluxes calculated from VAdiff; therefore, negative values indicate net removal, whereas positive values indicate net release. Some corrections to published data may apply to HEP_{NF}, TSP_{NF}, and MG_{NF} (refer to the text).² n = number of treatment means; Min. = minimum; Max. = maximum.³ Std Mpf = mammary plasma flow standardized across studies (refer to the text and Equation [2]).[‡] Mean is not different ($P > 0.10$) from 0.

Table 6. Estimation of the EndoSupply of EAA (g/d) for cows with DIM 1–30 in the selected dataset^{1,2}

Variable	n	Mean	Median	SD	Min.	Max.
His	19	14.7	14.5	5.30	1.3	23.0
Ile	24	14.1	11.6	12.37	0.0	39.6
Leu	24	23.0	15.3	23.29	0.0	67.6
Lys	20	28.0	28.6	15.85	0.0	65.9
Met	22	10.2	10.2	6.89	0.0	22.7
Phe	23	21.2	19.9	11.70	1.6	43.2
Thr	21	17.4	16.8	9.64	0.0	33.9
Trp	20	10.7	10.7	4.20	0.0	15.5
Val	23	13.3	12.2	12.69	0.0	42.0

¹EndoSupply was assumed to be 0 for cows later in lactation (DIM 31–305).

²n = number of treatment means; Min. = minimum (set to 0 if negative); Max. = maximum.

Potential Catabolism of EAA in Tissues

The potential catabolism values of EAA by the PDV, the liver, and the postsplanchnic tissues are detailed in Tables 7 and 8.

The PDV_{CATA}. As reported in Martineau et al. (2025), PDV_{CATA} can be used as a proxy of the oxidation of EAA by the PDV. Note that because we are using mEAA as the starting point, the first-pass catabolism is accounted for in our approach. Positive values indicate that estimated mEAA minus their excretion in MFP are, on average, greater than observed PDV_{NF}, which suggests a potential catabolism or oxidation of these EAA by the PDV. In both stages of lactation, PDV_{CATA} of the BCAA and Thr were positive and different from 0 ($P \leq 0.04$), indicating potential catabolism by the PDV. The PDV_{CATA} of His, Met, and Trp were either not different from 0 ($P \geq 0.16$) or even negative ($P \leq 0.03$). Contrary to these observations, PDV_{CATA} of Lys tended ($P = 0.09$) and that of Phe was positive ($P = 0.005$) for cows in early lactation but not different from 0 ($P \geq 0.42$) for cows in later lactation. In addition, the PDV_{CATA} of Leu, Met, and Phe were greater ($P \leq 0.02$), and those of Ile, Lys, and Thr tended ($P \leq 0.08$) to be greater for cows in early versus later lactation (Table 7).

The HEP_{CATA}. Contrary to PDV_{CATA}, only the BCAA did not show potential catabolism by the liver (no difference from 0 or negative value; Table 7). In contrast, HEP_{CATA} of group 1 AA, Lys, and Thr were positive and different from 0 ($P \leq 0.008$). The HEP_{CATA} of Met and Phe were smaller ($P \leq 0.01$) for cows in early versus later lactation. Although negative, the HEP_{CATA} of Ile and Leu tended ($P \leq 0.09$) to be greater for cows in early versus later lactation. Finally, the HEP_{CATA} of His, Lys, Thr, Trp, and Val did not differ ($P \geq 0.18$) between stages of lactation.

The PostTSP_{CATA}. For cows in early lactation, the PostTSP_{CATA} were negative ($P \leq 0.01$) for all EAA except for the BCAA, for which they were not different from 0 ($P \geq 0.38$) or tended ($P = 0.093$) to be positive

(Ile; Table 7). In contrast, for cows in later lactation, the PostTSP_{CATA} were either negative or not different from 0 for group 1 AA but positive ($P \leq 0.02$) for the BCAA, Lys, and Thr. On average, the PostTSP_{CATA} of all EAA were smaller ($P \leq 0.03$) for cows in early versus later lactation.

It must be acknowledged that a null, or even, negative PostTSP_{CATA} does not exclude potential catabolism occurring postliver because there might be additional postsplanchnic supply of EAA (from endogenous tissue mobilization or redistribution). Therefore, PostTSP_{CATA} was separated into PERI_{CATA} and MG_{CATA} to provide further examination on sites of potential catabolism, despite a global indication of no catabolism by postsplanchnic tissues. The PERI_{CATA} and MG_{CATA} are reported in Table 8, which includes a section of data for studies that reported MG_{NF} data and another section also including reported data and data calculated using a proxy of std MG_{NF} (see Equations [9], [10], and [11]). The PERI_{CATA} and MG_{CATA} did not differ significantly ($P \geq 0.08$) in the 2 sections, and therefore, Results and Discussion will focus on the section including reported plus calculated data.

The PERI_{CATA}. For cows in early lactation, the PERI_{CATA} of Val tended ($P = 0.06$) and those of the other EAA were negative ($P \leq 0.002$). In contrast, for cows in later lactation, the PERI_{CATA} of the BCAA and Lys were positive ($P < 0.001$), those of His, Met, and Phe were negative ($P \leq 0.04$), and those of Thr and Trp were not different from 0 ($P \geq 0.90$). Nevertheless, the PERI_{CATA} of all EAA were smaller ($P \leq 0.008$) for cows in early versus later lactation.

The MG_{CATA}. Overall, MG_{CATA} of all EAA were positive ($P \leq 0.004$) except that of Met ($P = 0.94$) for cows in later lactation. The MG_{CATA} of group 1 AA and Thr were greater ($P \leq 0.02$) for cows in early versus later lactation and the MG_{CATA} of Val ($P < 0.001$) went in the opposite direction with stage of lactation. The MG_{CATA} of Ile, Leu, and Lys did not differ ($P \geq 0.33$) between the 2 stages of lactation.

Table 7. Potential tissue catabolism of EAA (g/d) for the 2 stages of lactation in the selected dataset¹

Variable	Cows with DIM 1–30						Cows with DIM 31–305					
	n	Mean	Median	SD	Min.	Max.	n	Mean	Median	SD	Min.	Max.
PDV_{CATA}												
His	19	2.5 [‡]	1.5	7.59	−10.9	24.7	26	0.8 [‡]	0.9	4.05	−9.1	7.2
Ile	24	18.0 ^x	16.5	13.64	−8.3	47.0	25	10.5 ^y	13.0	14.83	−29.7	31.5
Leu	24	26.4 ^a	20.1	21.58	−2.4	66.0	26	10.6 ^b	15.4	25.11	−47.8	46.2
Lys	20	6.4 ^x	3.7	15.99	−20.7	38.7	27	−2.7 ^y	−3.1	16.99	−37.4	24.6
Met	22	2.7 ^A	3.3	8.89	−12.1	22.8	22	−3.5 ^B	−4.6	5.48	−15.7	5.1
Phe	23	9.0 ^A	13.5	13.77	−13.2	30.4	23	−2.0 ^B	−3.1	14.08	−25.0	21.5
Thr	21	19.8 ^x	17.6	10.88	4.3	42.0	17	14.0 ^y	16.3	9.13	0.5	32.0
Trp	20	−1.5 [‡]	0.4	6.99	−14.2	11.3	15	−6.6	−3.1	10.79	−37.7	4.8
Val	23	20.9	20.5	14.24	−10.9	51.3	25	13.0	21.6	24.16	−49.7	52.1
HEP_{CATA}												
His	19	10.8	10.0	7.24	−1.6	24.1	26	10.0	10.4	6.94	−2.0	27.1
Ile	24	−0.3 ^x	0.0	11.04	−28.4	20.3	25	−5.2 ^y	−7.0	9.09	−16.8	24.6
Leu	24	−1.7 ^x	−2.4	11.57	−29.6	16.5	26	−7.2 ^y	−8.5	10.16	−22.0	18.0
Lys	20	5.8	5.4	7.02	−6.5	20.9	27	5.5	6.0	9.16	−12.7	25.2
Met	22	9.7 ^B	8.4	5.40	0.4	20.7	22	16.1 ^A	16.5	6.52	2.1	24.8
Phe	23	39.5 ^B	40.2	15.12	17.5	70.8	23	53.0 ^A	47.9	19.25	16.8	89.3
Thr	21	18.9	19.8	10.34	0.5	39.8	17	23.2	21.2	11.62	−0.5	45.0
Trp	20	7.8	5.3	6.37	−1.3	18.8	15	10.3	10.3	4.48	2.4	17.8
Val	23	−2.6 [‡]	−2.3	13.16	−31.6	25.6	25	−3.9 [‡]	−8.8	13.51	−23.0	34.7
PostTSP_{CATA}												
His	19	−12.5 ^B	−12.0	5.51	−20.8	1.1	26	−1.3 ^A	−1.6	4.19	−10.3	6.4
Ile	24	5.4 ^B	5.3	14.97	−17.1	45.0	25	37.3 ^A	35.1	16.69	9.7	74.1
Leu	24	2.0 ^B	3.5	26.72	−46.2	51.5	26	48.9 ^A	44.9	25.99	11.4	101.8
Lys	20	−11.3 ^B	−9.6	18.33	−49.1	24.8	27	27.4 ^A	27.4	15.50	1.2	65.4
Met	22	−8.5 ^B	−8.6	6.80	−21.1	3.8	22	−1.6 ^A	−1.8	2.68	−7.5	4.8
Phe	23	−13.8 ^B	−11.6	10.96	−35.6	4.4	23	0.7 ^A	−0.2	4.19	−6.8	9.6
Thr	21	−12.4 ^B	−12.6	10.27	−28.6	11.9	17	3.4 ^A	4.3	5.15	−6.2	9.6
Trp	20	−5.9 ^b	−6.6	4.00	−10.5	5.4	15	2.0 ^y	−0.2	11.89	−8.1	41.9
Val	23	−3.9 ^B	−7.5	21.29	−44.0	46.0	25	31.3 ^A	30.1	21.48	2.7	75.4

^{a,b}Within rows, values with different superscripts differ ($P \leq 0.05$).^{x,y}Within rows, values with different superscripts tended ($P \leq 0.10$) to differ.^{A,B}Within rows, values with different superscripts differ ($P \leq 0.01$).¹n = number of treatment means; Min. = minimum; Max. = maximum.[‡]Mean is not different ($P > 0.10$) from 0.

Ratio U:O

Except for the U:O ratios of Met ($P \geq 0.35$; both stages of lactation) and Val ($P = 0.14$; cows in early lactation), the U:O of the other EAA were different from 1 ($P \leq 0.01$; Table 9). The U:O of Ile and Val were smaller ($P \leq 0.003$) for cows in early versus later lactation, in contrast to the U:O of Phe and Trp tending ($P \leq 0.09$) to be greater for cows in early versus later lactation. The U:O of His, Leu, Lys, Met, and Thr did not differ ($P \geq 0.31$) between stages of lactation.

Relationships Between NASEM Efficiencies and Measured Efficiencies

The results on the relationships between NASEM_{EffU} and Measured_{EffU} of each EAA are reported in Tables 10 and 11 for cows in early and later lactation, respectively. The corresponding relationships are depicted in Figures 3 and 4. Note that Measured_{EffU} were calculated

differently for group 1 AA, the BCAA, Lys, and Thr based on the significance of tissue catabolism.

Fit Statistics for Group 1 AA. For cows in early lactation, relationships between Measured_{EffU} and NASEM_{EffU} for group 1 AA are reported in Table 10 and depicted in Figure 3 (top 4 panels). The fit statistics were poor for His, Met, and Phe, as indicated by low CCC ($\leq 14\%$), and high RMSPE ($\geq 12\%$), RPE ($\geq 14\%$), and RSR ($\geq 1.26\%/%$) values. The latter (RSR = RMSPE $\div$ SD of Measured_{EffU}) was large ($\geq 4.14\%/%$) for Met and Phe, indicating that model performance was poor (Moriassi et al., 2007). Upon closer scrutiny, the SD for Measured_{EffU} was smaller for Met and Phe compared with that of His and Trp, with ranges averaging 52%, 9%, 23%, and 71% of the mean Measured_{EffU} for His, Met, Phe, and Trp, respectively. This indicates a limitation for Measured_{EffU} of Met and Phe compared with the corresponding NASEM_{EffU}. The ECT and ER were $\geq 68\%$ of MSPE for Phe and Met, respectively. Accordingly, the mean bias was positive and beyond acceptable limits for Phe (Measure-

Table 8. Potential PERI_{CATA} and MG_{CATA} (g/d) for the 2 stages of lactation in the selected dataset¹

Variable	Cows with DIM 1–30						Cows with DIM 31–305					
	n	Mean	Median	SD	Min.	Max.	n	Mean	Median	SD	Min.	Max.
PERI_{CATA}												
Reported												
His	7	−15.8 ^B	−14.3	3.22	−20.4	−12.1	24	−2.7 ^A	−3.1	4.76	−16.0	7.3
Ile	8	−16.1 ^B	−12.7	13.88	−39.6	2.7	21	17.8 ^A	13.6	18.29	−7.7	61.0
Leu	8	−22.4 ^B	−16.5	22.50	−60.8	3.2	20	27.3 ^A	22.0	23.17	0.5	80.3
Lys	7	−18.3 ^B	−15.7	9.70	−33.6	−6.8	21	12.0 ^A	11.7	15.12	−7.1	60.1
Met	7	−7.0 ^y	−4.6	6.44	−17.0	0.0	18	−1.2 ^{x‡}	−1.1	3.14	−5.8	5.2
Phe	7	−19.7 ^B	−19.9	9.64	−34.5	−7.2	18	−3.4 ^{A‡}	−3.5	3.73	−10.1	4.7
Thr	8	−19.2 ^B	−16.6	8.36	−33.9	−9.2	17	0.2 ^{A‡}	1.0	6.14	−11.9	10.8
Trp	8	−8.7 ^b	−8.5	6.27	−14.8	3.4	15	0.1 ^{a‡}	−3.9	13.16	−9.9	44.2
Val	7	−19.4 ^B	−19.2	11.12	−42.0	−7.1	21	15.5 ^A	14.7	21.07	−12.4	60.1
Reported + calculated ²												
His	19	−14.7 ^B	−14.5	5.30	−23.0	−1.3	26	−2.5 ^A	−3.1	4.72	−16.0	7.3
Ile	24	−12.1 ^B	−11.6	15.95	−39.6	30.8	25	19.3 ^A	16.2	17.33	−7.7	61.0
Leu	24	−19.9 ^B	−15.3	27.44	−67.6	27.9	26	28.6 ^A	22.5	22.55	0.5	80.3
Lys	20	−27.8 ^B	−28.6	16.14	−65.9	2.9	27	13.7 ^A	12.0	15.78	−8.0	60.1
Met	22	−10.2 ^B	−10.2	6.94	−22.7	0.6	22	−1.6 ^A	−1.6	3.37	−7.5	5.2
Phe	23	−21.2 ^B	−19.9	11.70	−43.2	−1.6	23	−3.1 ^A	−3.1	3.50	−10.1	4.7
Thr	21	−17.0 ^B	−16.8	10.42	−33.9	7.5	17	0.2 ^{A‡}	1.0	6.14	−11.9	10.8
Trp	20	−10.5 ^B	−10.7	4.70	−15.5	3.4	15	0.1 ^{a‡}	−3.9	13.16	−9.9	44.2
Val	23	−8.5 ^B	−12.2	20.27	−42.0	40.8	25	16.3 ^A	14.9	19.55	−12.4	60.1
MG_{CATA}												
Reported												
His	7	2.2	2.5	1.53	0.2	4.3	24	1.2	0.7	2.20	−2.1	5.8
Ile	8	18.8	17.2	4.58	13.5	25.4	21	18.6	17.6	6.46	7.5	30.7
Leu	8	23.7	22.7	10.51	10.0	38.7	20	20.9	19.3	10.49	4.3	49.6
Lys	7	16.9	18.0	9.86	0.7	31.6	21	14.4	18.5	17.62	−18.7	43.9
Met	7	1.9 [‡]	2.9	4.42	−4.8	7.6	18	0.0 [‡]	0.0	2.11	−3.1	3.9
Phe	7	7.8 ^A	8.9	3.50	3.1	13.4	22	3.2 ^B	3.1	2.62	0.0	8.0
Thr	8	4.9	4.9	1.76	2.2	7.6	17	3.2	3.6	2.26	−1.2	8.3
Trp	8	4.9 ^a	5.3	2.96	1.0	9.7	15	1.8 ^b	1.9	2.56	−2.3	7.4
Val	7	4.5 ^{B‡}	3.9	7.27	−3.0	16.5	21	15.4 ^A	15.7	7.36	0.2	27.5
Reported + calculated ²												
His	19	2.2 ^a	2.2	0.90	0.2	4.3	30	1.2 ^b	0.8	1.96	−2.1	5.8
Ile	24	17.5	17.1	3.02	13.5	25.4	31	18.2	18.0	5.56	7.5	30.7
Leu	24	21.9	21.3	6.19	10.0	38.7	30	20.6	20.1	8.73	4.3	49.6
Lys	20	16.5	16.4	5.65	0.7	31.6	31	13.7	13.6	14.48	−18.7	43.9
Met	22	1.7 ^A	1.6	2.37	−4.8	7.6	26	0.0 ^{B‡}	0.0	1.74	−3.1	3.9
Phe ³	23	7.4 ^A	7.3	1.95	3.1	13.4	27	3.3 ^{B‡}	3.7	2.38	0.0	8.0
Thr	21	4.6 ^A	4.5	1.09	2.2	7.6	23	3.2 ^B	3.0	1.93	−1.2	8.3
Trp	23	4.5 ^A	4.4	1.76	1.0	9.7	27	1.7 ^B	1.7	1.89	−2.3	7.4
Val	23	4.6 ^B	4.7	3.82	−3.0	16.5	31	15.1 ^A	15.7	6.18	0.2	27.5

^{a,b}Within rows, values with different superscripts differ ($P \leq 0.05$).^{x,y}Within rows, values with different superscripts tended ($P \leq 0.10$) to differ.^{A,B}Within rows, values with different superscripts differ ($P \leq 0.01$).¹n = number of treatment means; Min. = minimum; Max. = maximum.²Calculated using estimates of MG_{NF} based on MilkAA_{MG} times its average U:O ratio within each stage of lactation (refer to Table 9).³Based on the calculation of Mpf, the amount assessed to Phe indicates the excess Phe uptake needed to cover the lack of Tyr taken up by the mammary gland to account for Tyr secreted in milk proteins synthesized in the mammary gland.[‡]Mean is not different ($P > 0.10$) from 0.

d_{EffU} > NASEM_{EffU}). For Met and Phe, the slopes between Measured_{EffU} and NASEM_{EffU} were close to 0%/ % leading to linear biases beyond acceptable limits.

For cows in later lactation, the relationships between Measured_{EffU} and NASEM_{EffU} for group 1 AA are reported in Table 11 and depicted Figure 4 (top 4 panels). Contrary to cows in early lactation, Trp had the worst CCC (−7%), whereas the other group 1 AA had moderate

CCC (50%–78%), with small averaged RMSPE (10%), RPE (15%), and RSR (1.04%/ %) values. The ECT was ≥61% of MSPE for Met and Trp, but ER was ≤23% of MSPE for the 4 EAA. Accordingly, mean biases were negative and beyond acceptable limits for Met and Trp (Measured_{EffU} < NASEM_{EffU}). The slopes between Measured_{EffU} and NASEM_{EffU} of His and Trp were, respectively, 0.55%/ % and −0.20%/ %, leading to linear biases

Table 9. Ratio U:O of EAA for the 2 stages of lactation included in the selected dataset^{1,2}

Variable	Cows with DIM 1–30						Cows with DIM 31–305					
	n	Mean	Median	SD	Min.	Max.	n	Mean	Median	SD	Min.	Max.
His	7	1.07	1.07	0.048	1.01	1.13	24	1.04	1.03	0.075	0.90	1.19
Ile	8	1.24 ^B	1.23	0.039	1.19	1.31	21	1.31 ^A	1.30	0.080	1.15	1.45
Leu	8	1.18	1.18	0.066	1.09	1.28	20	1.21	1.19	0.108	1.05	1.50
Lys	7	1.16	1.18	0.096	1.01	1.32	21	1.15	1.21	0.217	0.73	1.47
Met	7	1.05 [‡]	1.07	0.119	0.86	1.20	18	1.00 [‡]	1.00	0.076	0.90	1.16
Phe	7	1.12 ^x	1.13	0.052	1.05	1.20	18	1.08 ^y	1.08	0.049	1.00	1.17
Thr	8	1.09	1.09	0.031	1.04	1.13	17	1.07	1.07	0.047	0.97	1.16
Trp	8	1.24 ^x	1.26	0.134	1.06	1.44	15	1.12 ^y	1.15	0.157	0.82	1.41
Val	7	1.06 ^{B,‡}	1.05	0.092	0.97	1.22	21	1.24 ^A	1.26	0.111	1.00	1.42

^{x,y}Within rows, values with different superscripts tended ($P \leq 0.10$) to differ.

^{A,B}Within rows, values with different superscripts differ ($P \leq 0.01$).

¹Data computed for cows in early (DIM 1–30) and later lactation (DIM 31–305). Refer to the text for details on calculation.

²n = number of treatment means; Min. = minimum; Max. = maximum.

[‡]Mean is not different ($P > 0.10$) from 1.

barely beyond acceptable limits for His at 10.1%_{obs.mean} but compromised for Trp at 22.1%_{obs.mean}. In contrast to cows in early lactation, the SD of Measured_{EffU} was similar across group 1 AA, with ranges averaging 58%, 52%, 59%, and 47% of the mean Measured_{EffU} for His, Met, Phe, and Trp, respectively.

Fit Statistics for the BCAA. For cows in early lactation, the relationships between Measured_{EffU} and NASEM_{EffU} of the BCAA are reported in Table 10 and shown in Figure 3 (bottom panels). Isoleucine and Leu had moderate CCC ($\geq 52\%$), in contrast to Val, which had a poor CCC (23%) associated with the highest ER and RSR among the BCAA and a slope of 0.28%/ % compared with $\geq 0.69\%$ for Ile and Leu. Nonetheless, mean and linear biases were within acceptable limits for the BCAA. These relationships are depicted in Figure 3 which also displays the better goodness-of-fit for the BCAA compared with that of group 1 AA.

For cows in later lactation, the relationships between Measured_{EffU} and NASEM_{EffU} of the BCAA are reported in Table 11 and shown in Figure 4 (bottom panels). The fit statistics were good for the BCAA, as indicated by high CCC ($\geq 71\%$), and small RMSPE ($\leq 6\%$), RPE ($\leq 10\%$), and RSR ($\leq 0.81\%$) values. The decomposed MSPE was $\geq 47\%$ of MSPE for Ile and Leu (ECT), but ER was $\leq 5\%$ of MSPE for the BCAA. Accordingly, all mean biases were slightly positive (Measured_{EffU} < NASEM_{EffU}) but within acceptable limits, as were also the linear biases.

Fit Statistics for Lys. For cows in early lactation, the relationship between Measured_{EffU} and NASEM_{EffU} of Lys is reported in Table 10 and shown in Figure 3 (bottom right panel). Lysine had a poor CCC (29%), small RMSPE and RPE ($\geq 8\%$) but high RSR (2.73%/ %) values suggesting poor model performance. Similar to Met and Phe, the SD for Measured_{EffU} of Lys was small (2.8%),

with the range of Lys values averaging 13% of the mean Measured_{EffU}. This indicates a limitation for Measured_{EffU} of Lys (along with Met and Phe) compared with their corresponding NASEM_{EffU}. The decomposed MSPE of Lys were 27% and 64% of MSPE for ECT and ER, respectively. Accordingly, the mean bias was slightly positive (Measured_{EffU} < NASEM_{EffU}) but within acceptable limits. In line with the high ER, the slope between Measured_{EffU} and NASEM_{EffU} was 0.20%/ %, leading to a linear bias beyond acceptable limits.

For cows in later lactation, the relationship between Measured_{EffU} and NASEM_{EffU} of Lys is reported in Table 11 and depicted in Figure 4 (bottom right panel). The CCC, RMSPE, RPE, RSR, ECT, and ER values all indicated a good degree of agreement between EffU. Mean and linear biases were within acceptable limits despite the slope between Measured_{EffU} and NASEM_{EffU} of Lys (0.74%/ %) being smaller compared with the BCAA (0.88%/ %, on average).

Fit Statistics for Thr. For cows in later lactation, the relationship between Measured_{EffU} and NASEM_{EffU} of Thr is reported in Table 11 and depicted in Figure 4 (bottom right panel). The CCC was the lowest of studied EAA, with a slope between Measured_{EffU} and NASEM_{EffU} (0.38%/ %) being the smallest of group 2 AA, leading to a linear bias, although the latter remained within acceptable limits. There was no mean bias.

DISCUSSION

Because the dataset used was limited to studies that measured splanchnic net fluxes, it constituted only a fraction of the studies reporting mammary net fluxes in dairy cows. This is why the separation of PostTSP_{CATA} between PERI_{CATA} and MG_{CATA} has to be interpreted with caution, especially for cows in early lactation (n = 8). Nevertheless,

Table 10. Performance of NASEM_{EffU} to predict Measured_{EffU} of EAA (%; adjusted for the effect of study) for cows in early lactation (DIM 1–30)^{1,2}

Variable	Group 1 AA				BCAA			
	His	Met	Phe	Trp	Ile	Leu	Val	Lys
n	19	22	23	20	24	24	23	20
NASEM _{EffU}								
Mean	74	75	58	70	70	76	79	81
SD	6.9	10.6	6.7	10.1	5.8	9.1	6.3	7.9
Min.	58	55	46	55	58	60	65	65
Max.	85	98	70	97	79	93	93	92
Measured _{EffU}								
Mean	82	82	70	75	72	76	80	85
SD	11.0	2.2	3.5	13.2	9.4	8.0	7.6	2.8
Min.	60	79	61	59	51	59	69	80
Max.	103	86	77	112	92	91	101	91
Fit statistics								
CCC (%)	14	11	0	84	52	78	23	29
RMSPE (%)	14	12	14	7	8	6	9	8
RPE (% _{obs.mean})	17	14	21	9	11	7	11	9
ECT (% MSPE)	33	29	73	60	4	0	2	27
ER (% MSPE)	10	68	21	14	0	23	27	64
RSR (%/%)	1.26	5.33	4.14	0.52	0.81	0.70	1.13	2.73
Mean bias ³ (%)	7.9**	6.2**	12.3***	5.3***	1.4	0.3	1.1	4.0*
Mean bias ³ (% _{obs.mean})	9.7	7.7	17.5	7.1	2.0	0.4	1.3	4.7
Slope ⁴ (%/%)	0.35†	0.08***	−0.01***	1.26**	0.95	0.69*	0.28**	0.20***
SE _{Slope} ⁴ (%/%)	0.379	0.043	0.113	0.084	0.278	0.118	0.255	0.070
Linear bias ^{3,5} (%)	4.9	9.2	10.9	1.8	0.4	4.0	5.6	9.7
Linear bias ³ (% _{obs.mean})	6.0	11.3	15.6	2.3	0.6	5.2	7.1	11.3

¹Measured_{EffU} were adjusted for the effect of experiment, as illustrated in St-Pierre (2001), to reflect the ability of NASEM to predict changes in measured EffU of each EAA within the experiment.

²n = number of treatment means; Min. = minimum; Max. = maximum.

³Mean and linear biases in boldface type are beyond acceptable limits ($\geq 10.0\%$ _{obs.mean}). The *P*-value of the mean bias is denoted by a symbol.

⁴The coefficient and SE of the slope are from the simple linear regression of adjusted Measured_{EffU} values versus predicted NASEM_{EffU} values. The *P*-value of the slope being different from 1 is denoted by a symbol.

⁵The linear bias is the absolute difference between biases computed for first and third quantiles of NASEM_{EffU} values.

†*P* ≤ 0.10; **P* ≤ 0.05; ***P* ≤ 0.01; ****P* ≤ 0.001.

the U:O of cows in later lactation globally followed the patterns proposed by Mephram (1982) and Lapierre et al. (2012) for groups 1 and 2 AA. Indeed, the U:O of Phe was greater than 1 to account for the mammary synthesis of Tyr. To simplify and structure the discussion, the nominal categories of group 1 and 2 AA will be retained, although, as will be seen, the integration of whole-body metabolism rather than only mammary metabolism revealed different metabolic patterns within group 2 AA.

It was decided to present the data on Trp as a basis for future research, but results will not be discussed for the following reasons. First, few transorgan Trp net fluxes were available for cows in later lactation. Second, the metabolizable supply of Trp needed to be treated with caution because although postruminal outflows were predicted by NASEM, Trp residuals were not evaluated (see Table 20-14 in NASEM, 2021); indeed, Martineau et al. (2024) mentioned that outflows of Trp were reported in only 27 studies. Third, plasma Trp is not an easy AA to analyze (small peak, coming out at the end of the chromatogram) and has the smallest net flux compared with the other AA, resulting in low expected

VAdiff, often at the limit of the sensitivity of analyses. The accumulation of these biological and technical constraints could hinder identification of response patterns for Trp. The fact that the U:O of Trp largely exceeded 1 in this database suggests that the Trp metabolic pattern might not be considered to follow group 1 AA. As stated previously, however, this database was not intended to study mammary metabolism, and the pattern of mammary Trp metabolism needs to be studied with a more complete dataset using more rigorous analytical procedures.

Group 1 AA

Cows in Early Lactation. For cows in early lactation, the liver removal of group 1 AA was important, averaging 37%, 29%, and 48% of PDV_{NF} for His, Met and Phe, respectively. This resulted in a postliver supply of group 1 AA insufficient to cover MilkAA_{MG}. Indeed, the ratios MilkAA_{MG}:TSP_{NF} of group 1 AA being greater than 1 indicated that, during this period, the cows had to rely to a significant degree on EndoSupply. This was

Table 11. Performance of NASEM_{EffU} to predict Measured_{EffU} of EAA (%; adjusted for the effect of study) for cows in later lactation (DIM 31–305)^{1,2}

Variable	Group 1 AA				BCAA				
	His	Met	Phe	Trp	Ile	Leu	Val	Lys	Thr
n	26	22	23	15	25	26	25	27	17
NASEM _{EffU}									
Mean	81	76	56	79	63	71	67	78	63
SD	11.3	10.4	9.2	9.1	7.4	9.5	8.2	10.6	5.8
Min.	59	56	36	62	52	57	54	59	52
Max.	102	91	69	92	77	88	86	105	73
Measured _{EffU}									
Mean	79	63	53	62	59	67	65	75	63
SD	11.7	9.3	7.6	9.2	7.3	8.8	9.4	10.3	4.5
Min.	56	45	37	46	44	52	48	53	50
Max.	102	78	68	75	72	84	84	90	69
Fit statistics									
CCC (%)	53	50	78	–7	71	89	79	72	48
RMSPE (%)	11	13	6	22	6	4	6	8	5
RPE (% _{obs.mean})	14	21	11	35	10	6	9	10	8
ECT (% MSPE)	2	94	36	61	47	60	18	17	0
ER (% MSPE)	20	1	20	23	5	5	1	12	46
RSR (%/%)	0.95	1.42	0.75	2.35	0.81	0.49	0.61	0.76	1.14
Mean bias ³ (%)	–1.7	–12.9***	–3.5**	–16.9***	–4.0***	–3.4***	–2.4*	–3.2*	0.2
Mean bias ³ (% _{obs.mean})	–2.1	–20.4	–6.5	–27.3	–6.9	–5.0	–3.8	–4.3	0.3
Slope ⁴ (%/%)	0.55*	0.85*	0.71**	–0.20***	0.81	0.89†	0.94	0.74*	0.38**
SE _{Slope} ⁴ (%/%)	0.179	0.060	0.092	0.277	0.117	0.056	0.133	0.127	0.174
Linear bias ^{3,5} (%)	7.9	2.7	4.2	13.7	1.9	1.8	0.7	4.0	4.8
Linear bias ³ (% _{obs.mean})	10.1	4.2	7.9	22.1	3.2	2.6	1.0	5.4	7.6

¹Measured_{EffU} were adjusted for the effect of experiment, as illustrated in St-Pierre (2001), to reflect the ability of NASEM to predict changes in measured EffU of each EAA within the experiment.

²n = number of treatment means; Min. = minimum; Max. = maximum.

³Mean and linear biases in boldface type are beyond acceptable limits ($\geq 10.0\%$ _{obs.mean}). The *P*-value of the mean bias is denoted by a symbol.

⁴The coefficient and SE of the slope are from the simple linear regression of adjusted Measured_{EffU} values versus predicted NASEM_{EffU} values. The *P*-value of the slope being different from 1 is denoted by a symbol.

⁵The linear bias is the absolute difference between biases computed for first and third quantiles of NASEM_{EffU} values.

†*P* ≤ 0.10; **P* ≤ 0.05; ***P* ≤ 0.01; ****P* ≤ 0.001.

further supported by the significant negative PERI_{CATA} calculated for this period.

The EndoSupply in early lactating cows represented 13% to 24% of the metabolizable supply of Ile, Leu, Lys, Met, Phe, Thr, and Val, and 34% of the metabolizable His. Indeed, the needed minimum EndoSupply of EAA would be 42%, 27%, and 23% of PDV_{NF} to cover MilkAA_{MG} for His, Met, and Phe, respectively. Endogenous sources of EAA would include uterine involution and depletion of body protein “reserve” for all EAA (NASEM, 2021) plus depletion of muscle pools of carnosine and blood hemoglobin for His (Lapierre et al., 2021). This observation agrees with cows being in negative N balance during the first 4 wk after calving (Tamminga et al., 1997; Daniel et al., 2022). A contribution of EAA from tissue mobilization was reported to occur in early lactating cows (Komaragiri and Erdman, 1997; Pedernera et al., 2008; Daniel et al., 2022).

In NASEM (2021), a supply originating from uterine involution in EAA supply is included during the first 21 d of lactation. Essential AA released by uterine involution would make a relatively small contribution to total EAA supply, with the highest contribution on d 1

at 202 g of TP per day (NASEM, 2021). This compares with the estimated 1.2 to 2.5 kg MP supply (Table 4). The relationship between PERI_{CATA} and DIM for group 1 AA would theoretically indicate the minimum amount of EAA needed to support MilkAA_{MG} from endogenous sources during that period, when summed for the first 30 DIM. These are approximate estimates, but the predicted EndoSupply required for the first 30 d postpartum averaged 379, 303, and 476 g of His, Met, and Phe, respectively. Using NASEM estimation of the contribution of uterine involution and AA composition of empty body, the latter would contribute between 17% and 20% of this EndoSupply, indicating an additional needed mobilization between 8.7 (Phe), 10.3 (His), and 10.7 (Met) kg of TP over the period, although the need of TP for His might be smaller due to the contribution of carnosine and hemoglobin. This amount is approximately twice the loss of CP predicted from the decrease in empty body carcass weight between calving and 5 wk of lactation (5.1 kg; Gibb et al., 1992) or the protein mobilization occurring during the first 4 wk of lactation (4.6 kg; Tamminga et al., 1997). Cows from the study of Gibb et al. (1992) produced, on average in their previous lactation, 4,807 kg

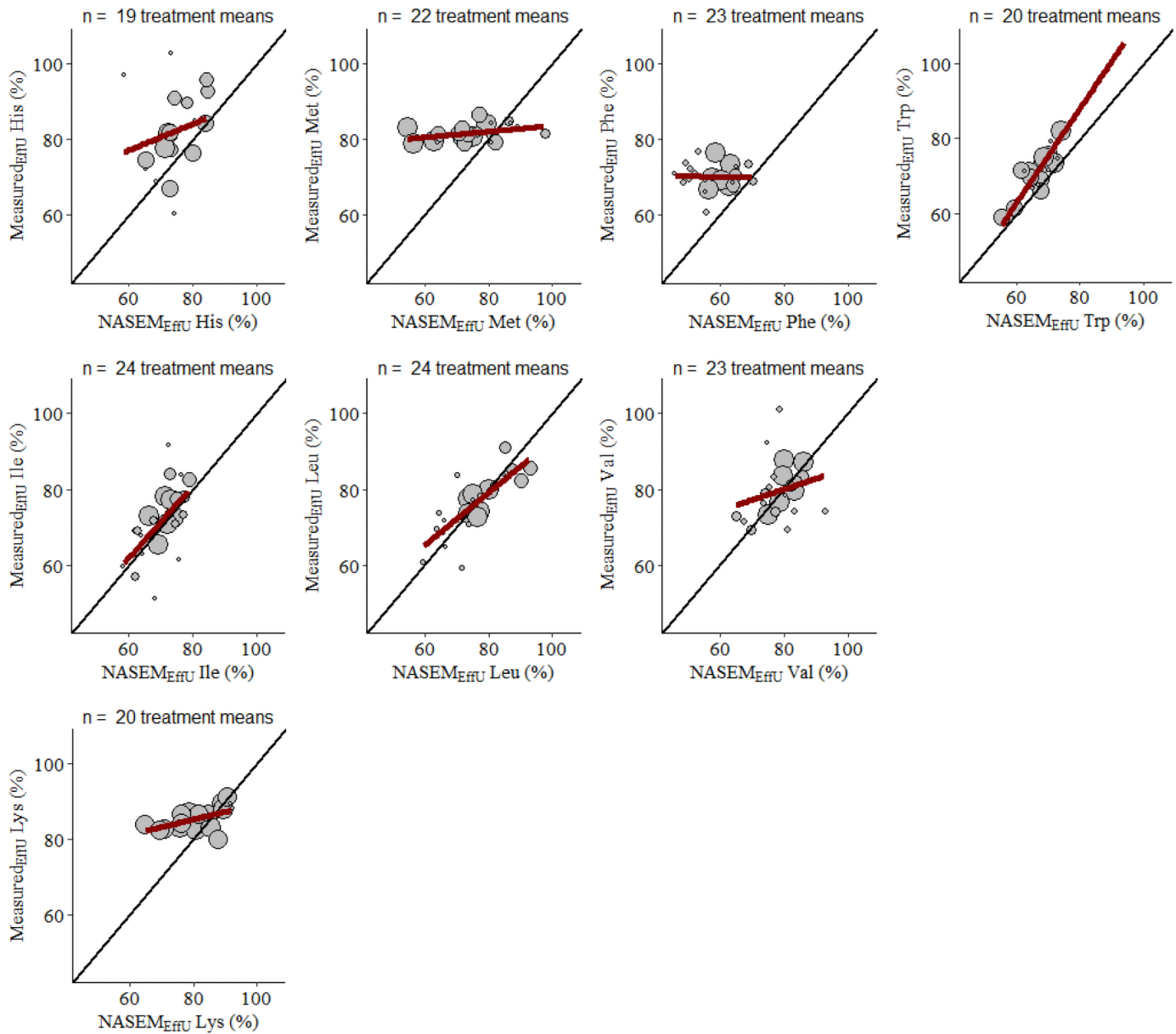


Figure 3. Relationships between $\text{Measured}_{\text{EffU}}$ and $\text{NASEM}_{\text{EffU}}$ of EAA for cows in early lactation (DIM 1–30; refer to Table 10). Solid bold red lines represent the best-fit simple regression line, and black lines represent the line of unity. Symbols are sized according to the standardized SEM.

for second-lactation cows and 5,763 kg for older cows, which may indicate lower producers than in the current study and therefore, a lower EAA demand to cover MPY.

Overall, net flux data for cows in early lactation confirmed that the main site of group 1 AA potential catabolism, not used to support protein export, is the liver. The negative $\text{PERI}_{\text{CATA}}$ clearly indicated a contribution from endogenous sources to support $\text{MilkAA}_{\text{MG}}$, which invalidated a direct calculation of EffU for cows in early lactation, as initially thought. Indeed, the presence of EndoSupply would decrease the $\text{NASEM}_{\text{EffU}}$ but increase $\text{Measured}_{\text{EffU}}$. We thought that estimating the contribution of EndoSupply

would help to overcome that pitfall. Despite this attempt, however, the relationships between $\text{Measured}_{\text{EffU}}$ and corresponding $\text{NASEM}_{\text{EffU}}$ remained too poor (CCC between 0% and 14%) to pursue an interpretation.

One intriguing point, however, was that the slopes between $\text{Measured}_{\text{EffU}}$ and $\text{NASEM}_{\text{EffU}}$ of Met and Phe were not significantly different from 0 ($P \geq 0.08$; Table 10; Figure 3, top panels), independently of the experimental conditions across studies. Furthermore, the average $\text{Measured}_{\text{EffU}}$ values of Met and Phe are similar to the maximal values observed in later lactation, with the exception of those under more extreme experimental conditions (e.g.,

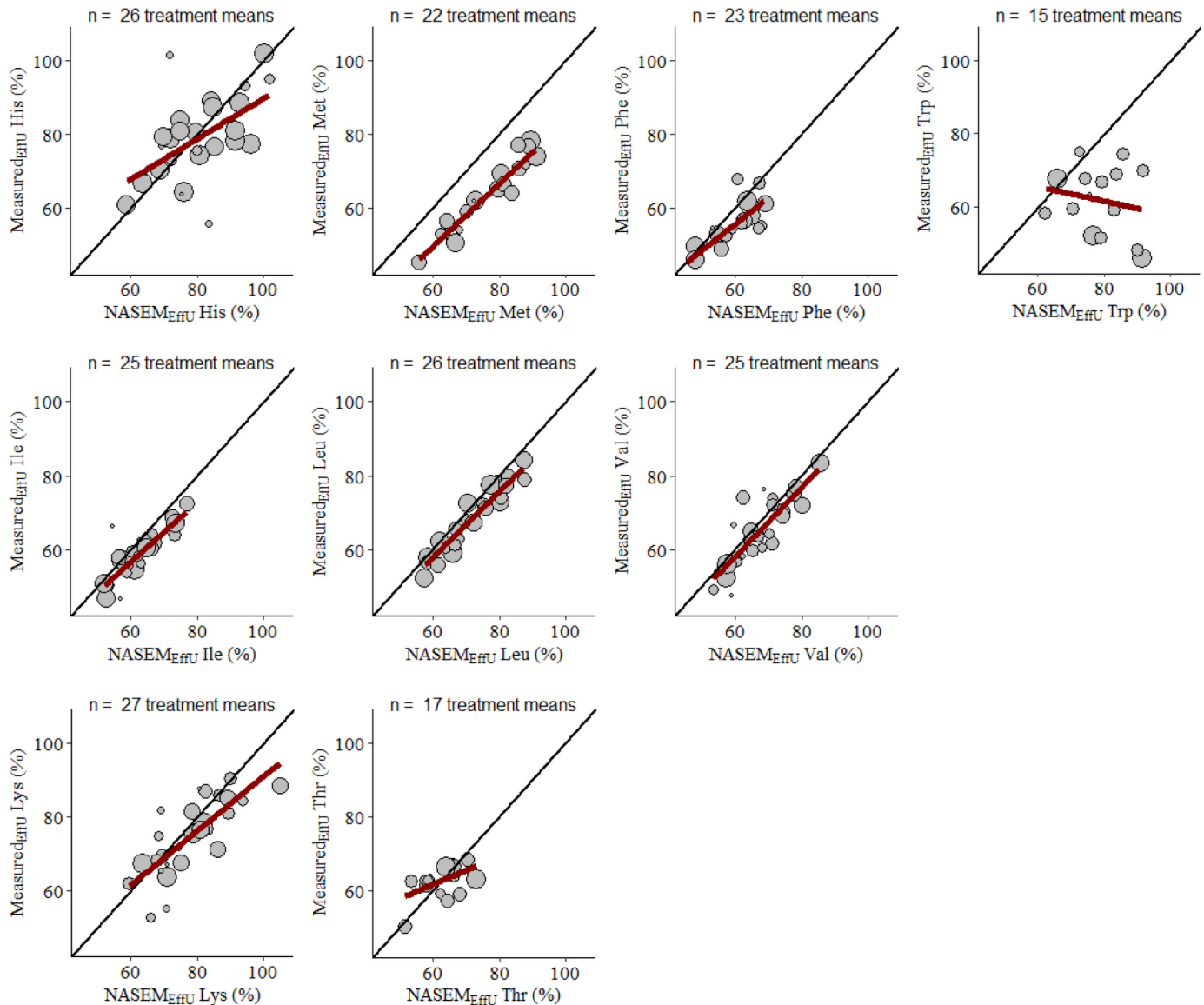


Figure 4. Relationships between Measured_{EffU} and NASEM_{EffU} of EAA for cows in later lactation (DIM 31–305; refer to Table 11). Solid bold red lines represent the best-fit simple regression line, and black lines represent the line of unity. Symbols are sized according to the standardized SEM.

very low CP diet, deletion studies, glucose infusion) for which the Measured_{EffU} were slightly higher. It was as if the cows were performing under biological constraints, at least for Met and Phe, during that critical postpartum period. As yet we have no plausible explanation for this very small variation in Measured_{EffU} of Met and Phe for cows in early lactation.

Cows in Later Lactation. For cows in later lactation, there was no estimated catabolism of group 1 EAA across the PDV. This was supported by observation in sheep (Lobley et al., 2003) with systemically supplied labeled Phe and Met, although there was slight oxidation of the latter. In the current analysis, liver removal was substantial, averaging 26%, 36%, and 49% of PDV_{NF} for

His, Met and Phe, respectively. During this period, the postliver supply of group 1 AA was sufficient to cover MilkAA_{MG}, although the ratios MilkAA_{MG}:TSP_{NF} may suggest a minor limitation in supply of Met because it was significantly different from 1 (Table 2). There was no potential catabolism of His, Met, and Phe by post-splanchnic tissues, indicating that most, if not all, of their removal, not covering export proteins, occurred across the liver. The net removal of EAA by the liver, however, does not necessarily imply that these EAA are directed toward oxidation. Indeed, the liver synthesizes several export proteins and, in dairy cows, synthesis of export proteins was responsible for 11% to 18% of Phe liver removal (Raggio et al., 2007). These uptakes do not

necessarily represent a net drain to the animal, however, because these export proteins remain “internal” (as opposed to those associated with export proteins used in NASEM). The EAA of these internal export proteins can be recovered following degradation at several tissues, including the liver (Maxwell et al., 1990). Due to the limited information on this topic, it was assumed that the liver removal of EAA was directed toward catabolism, except for the blood-borne proteins secreted in milk.

The mean and linear biases between $\text{Measured}_{\text{EffU}}$ and $\text{NASEM}_{\text{EffU}}$ for Phe remained within acceptable limits, with absolute estimates differing by 2.5%. The target efficiency of Phe is the lowest (0.60) among EAA in NASEM (2021) and this is reflected in the $\text{Measured}_{\text{EffU}}$ values reported in Tables 10 and 11. The reason for this low EffU for Phe is not clear and attempts to decrease Phe supply in deletion studies have shown that a high Phe supply (or low efficiency) is needed to support MPY (Doelman et al., 2015; Doepel et al., 2016). This high removal of Phe might be related to the synthesis of Tyr from Phe, especially if a relative deficiency in Tyr supply exists. The ratio of Tyr to Tyr+Phe in milk (0.53) is known to be higher than in microbial protein (0.48) and most feed ingredients (usually less than 0.45; NASEM, 2021). Any Tyr deficiency would need to be met by biosynthesis from Phe. In dairy cows, based on the respective whole-body rate of appearance of Phe and Tyr (Crompton et al., 2022) and 14% appearance of Tyr from labeled Phe (Lemosquet et al., 2010), this would give a minimum of 18 g/d conversion of Phe to Tyr. Based on the model of Crompton et al. (2014), approximately 16 g/d of the conversion of Phe to Tyr occurs in the liver. Lemosquet et al. (2010) reported that only 2.5% of Tyr in milk protein is synthesized within the mammary tissue, thus corroborating that the majority of Tyr synthesis would be hepatic-based. In the current dataset with cows later in lactation, the excess of mammary Phe uptake contributing to Tyr synthesis averaged 3.3 g/d, which would represent 6% of Tyr in milk protein secretion synthesized in the mammary gland. Based on rate of appearance data (Crompton et al., 2022), much of the Tyr synthesized in the liver is exported to the plasma and becomes available as a source for protein anabolism. In consequence, including the anabolic conversion of Phe to Tyr in the liver (Crompton et al. (2014) would increase the average $\text{Measured}_{\text{EffU}}$ of Phe from 53% to 73%, and thus within the range of the other EAA. However, such an approach would only apply where an EAA is a direct precursor of another AA incorporated into export proteins. The linear bias between $\text{Measured}_{\text{EffU}}$ and $\text{NASEM}_{\text{EffU}}$ of His averaged 10.1%_{obs.mean}, which was slightly above the threshold of 10.0% (Table 11). In Lapierre et al. (2021; Table 10), the $\text{Measured}_{\text{EffU}}$ of His were smaller compared with the EffU predicted using NRC (2001), but both EffU

were highly correlated (0.97) in that study with increasing dosage of His.

Unlike His and Phe, Met showed a compromised mean bias (−13% or −20%_{obs.mean}; Table 11 and Figure 4). This model inadequacy would support the idea that the metabolism of Met was not being captured adequately under the current assumptions. This would support the specific utilization of Met toward the synthesis of metabolites (1) not including other EAA in their composition, and (2) not related to export proteins. Analogous to the Phe and Tyr discussed previously, Met sulfur is required for the synthesis of Cys and, in sheep, a substantial portion of Cys in plasma albumin (5%–15%) and wool (15%–50%) is derived from Met (Pisulewski and Buttery, 1985). In milk protein, however, the demand for Cys relative to Met is low (only 0.93 g vs. 3.03 g per 100 g milk TP; see Table 6-2 in NASEM, 2021). Other than protein demand, Cys is required for synthesis of antioxidants such as glutathione and taurine, and half of the Cys required for glutathione synthesis was provided by the Met transsulfuration pathway in hepatocytes (Brosnan and Brosnan, 2006). Furthermore, Met is a key intermediate in the supply of methyl groups needed to support the hundred or so methylation reactions fundamental to mammalian metabolism. These key functions involve S-adenosyl methionine as an intermediate in the methionine cycle and, while much of the precursor Met can be regenerated, this is unlikely to be 100% efficient, even when allowance is made for synthesis of byproducts such as the antioxidants and polyamines. Indeed, the rate of this process cycle (Brosnan and Brosnan, 2006; Wilson et al., 2009) may lead to a potential drain on Met as part of maintaining the necessary supply of methyl donors. Although this phenomenon has been observed in dairy cows (Preynat et al., 2009), there is little information on the quantification of the net demand for Met for this type of metabolism in such animals. In addition, the supplementation of periparturient dairy cows with rumen-protected Met has been reported to help in reducing oxidative stress and inflammation (e.g., Osorio et al., 2014; Zhou et al., 2017). To equate $\text{NASEM}_{\text{EffU}}$ and $\text{Measured}_{\text{EffU}}$, the proportion of Met liver removal not directed toward oxidation should represent approximately 31% of Met liver removal, indicating that a substantial portion of Met removal would be used for these types of anabolic functions. Again, due to the variance associated with each measurement, caution needs to be taken in the interpretation of these absolute values.

Overall, for cows in established lactation, the net flux data of group 1 AA confirmed that the main site of catabolism, not used to support protein export, was the liver as was concluded for cows in early lactation. Considering the complexity of analyses and the numerous assumptions, the fit statistics between $\text{Measured}_{\text{EffU}}$

and $\text{NASEM}_{\text{EffU}}$ for His and Phe further corroborate the concepts behind $\text{NASEM}_{\text{EffU}}$ and the assumption that the liver removal of these EAA represents their degree of inefficiency. There is, however, an as yet unexplained negative mean bias ($\text{Measured}_{\text{EffU}} < \text{NASEM}_{\text{EffU}}$) for Met, indicating an usage not accounted for in the export proteins currently included by NASEM (see the numerator in Equation [12]).

Group 2 AA

For cows in early lactation, the ratios $\text{MilkAA}_{\text{MG}}:\text{TSP}_{\text{NF}}$ of the BCAA were not significantly different from 1. Although this suggested a postliver supply sufficient to support $\text{MilkAA}_{\text{MG}}$, the coverage of the BCAA requirements of the mammary gland were not fulfilled because the corresponding U:O varied from 1.06 to 1.24, indicating higher mammary gland uptake than $\text{MilkAA}_{\text{MG}}$ with occurrence of MG_{CATA} . This led to negative $\text{PERI}_{\text{CATA}}$ during that period, suggesting also that the EndoSupply of group 2 AA was needed to cover the mammary gland's requirements. Using the limited data on $\text{PERI}_{\text{CATA}}$ during this period ($n = 8$), the contribution from endogenous sources (excluding the uterine involution) would average between 5 and 11 kg of TP. Interestingly, these estimates are within the same range as those calculated above for group 1 AA.

For cows in later lactation, the averaged ratios $\text{MilkAA}_{\text{MG}}:\text{TSP}_{\text{NF}}$ of group 2 AA varied between 0.61 and 0.86, indicating that the TSP_{NF} were higher than the corresponding $\text{MilkAA}_{\text{MG}}$. This suggested substantial MG_{CATA} , in line with U:O consistently greater than 1. Classifying Lys in group 2 AA was based initially on mammary metabolism with U:O ratios above 1 (Mephram, 1982), and later reinforced by the limited Lys liver removal compared with group 1 AA (Lapierre et al., 2012). Although the BCAA and Lys, along with Thr, shared similar potential MG_{CATA} in the current study, they exhibited differences in their metabolic fates and, as such, will be discussed separately.

BCAA. The potential catabolism of the BCAA was restricted to the PDV and mammary gland for cows in early lactation, and PDV and PERI plus mammary gland for those in later lactation. For cows in early lactation, a net contribution from endogenous sources was needed to support mammary uptake, which was larger than $\text{MilkAA}_{\text{MG}}$, indicating that MG_{CATA} also occurred. Nonetheless, it cannot be ruled out that $\text{PERI}_{\text{CATA}}$ could occur, and if it did, the contribution from EndoSupply would need to be increased by the same amount, but determining if that happened was not possible. Within the limitation of the approach used, we had to assume that $\text{PERI}_{\text{CATA}}$ was not present in cows in early lactation. For cows in early lactation, the proportion of $\text{PostTSP}_{\text{CATA}}$ relative to total removal of the BCAA (i.e., $\text{MG}_{\text{CATA}} \div$

$[\text{PDV}_{\text{CATA}} + \text{MG}_{\text{CATA}}]$) averaged 38% (range = 18% to 49%) and this proportion increased to 77% (range = 71% to 82%) on average when computed for cows in later lactation (i.e., $[\text{PERI}_{\text{CATA}} + \text{MG}_{\text{CATA}}]/[\text{PDV}_{\text{CATA}} + \text{PERI}_{\text{CATA}} + \text{MG}_{\text{CATA}}]$), due to significant $\text{PERI}_{\text{CATA}}$ in later lactation. Overall, for cows in early lactation, the BCAA removal by the liver was not different from 0, in contrast to cows in later lactation, where a small net release of Ile and Leu was observed but is as yet unexplained. The oxidation of Leu has been reported in dairy cows across the PDV (Lapierre et al., 2002) and by the mammary gland (Bequette et al., 1996; Raggio et al., 2006). In sheep, Leu oxidation has been reported in the muscles (Harris et al., 1992). Because the BCAA share common enzymes for their degradation (branched-chain aminotransferase; branched-chain α -keto acid dehydrogenase complex), Ile and Val probably share a similar fate to Leu across the tissues. For cows in later lactation, the partition of catabolism between the tissues for Ile, Leu, and Val were, respectively, 0.22, 0.18, and 0.29 for the PDV; 0.37, 0.46, 0.35 for PERI; and 0.38, 0.35, and 0.34 for the mammary gland. Thus, for cows in established lactation, the majority of catabolism occurred postliver and was divided approximately equally between nonmammary peripheral tissues and the mammary gland.

Therefore, to estimate the inefficiency of the BCAA, PDV_{CATA} and MG_{CATA} were summed for cows in early lactation whereas $\text{PERI}_{\text{CATA}}$ was added for cows in later lactation. According to Equation [5], EndoSupply was included only for cows in early lactation. The reasonable CCC for Ile and Leu, and small mean or linear biases, all below 7.1%_{obs.mean}, indicated a good degree of agreement between the concepts underlying the calculation of the $\text{NASEM}_{\text{EffU}}$ and the assumptions made regarding the sites of potential catabolism of the BCAA and net flux measurements (see Figure 3). The relationships were better for cows in later lactation compared with cows in early lactation, especially for Val. Not all of excess BCAA from mammary removal is fully oxidized to provide energy; they can also act as an N-donors to support NEAA synthesis within the mammary gland, analogous to Lys (Lapierre et al., 2009), or the C-skeleton can be used for lipid or carbohydrate biosynthesis (Bequette et al., 2006; see review, Lapierre et al., 2012).

The dataset for the current analyses included not only cows at 2 stages of lactation but also under a variety of experimental treatments that could alter the total and tissue-specific catabolism. For example, the infusion of casein into the abomasum tended ($P \leq 0.08$) to increase the U:O of Ile and Leu (i.e., additional MG_{CATA} occurred; Raggio et al., 2006), but no effect was observed with energy (propionate) infusion. In a study in which splanchnic and mammary metabolism were quantified, an abomasal AA infusion comparable to the casein profile induced

similar increases in both MG_{CATA} and $PERI_{CATA}$ (Omphalius et al., 2020). In sheep, increased intake from $0.6\times$ to $1.8\times$ maintenance increased hindquarter oxidation 3-fold (Harris et al., 1992). Overall, the $Measured_{EffU}$ of the BCAA confirmed that the concepts underlying the estimation of $NASEM_{EffU}$ are biologically sound and that the catabolism of the BCAA is mainly confined to the PDV, peripheral tissues, and mammary gland, with no measurable removal by the liver.

Lysine. As mentioned earlier, Lys was initially classified in group 2 based on mammary gland dynamics ($U:O > 1$) but its metabolism differs from that of the BCAA. Based on PDV_{CATA} , Lys is not oxidized by the PDV, in agreement with the work of Martineau et al. (2025). Similarly, Lobley et al. (2003) reported no oxidation of Lys across the PDV in sheep using systemically infused labeled Lys. Although the HEP_{CATA} of Lys was relatively small (6 g/d), it was significantly different from 0 and represented, on average, 26% or 17% of total catabolism for cows in early versus later lactation, respectively. Therefore, to estimate the inefficiency of Lys, HEP_{CATA} and MG_{CATA} were summed for cows in early lactation, whereas $PERI_{CATA}$ was also added for cows in later lactation. For cows in early lactation, the linear bias was above acceptable limits. However, for cows in later lactation, the concepts underlying the estimation of $NASEM_{EffU}$ gave a good agreement, with mean and linear biases within acceptable limits, similar to those for the BCAA. The fractions of total catabolism observed in each tissue were 0.17, 0.45 and 0.38 for HEP_{CATA} , MG_{CATA} and $PERI_{CATA}$ respectively. These data confirm that the major sites of Lys oxidation in established lactation are the PERI and mammary gland, with less than 20% contributed by the liver. Similar to the BCAA, an increased supply of AA in general (Omphalius et al., 2020) or Lys alone (Lapierre et al. 2009) resulted in similar increases in both MG_{CATA} and $PERI_{CATA}$. As also stated for the BCAA, part of the excess mammary uptake of Lys can be used for anabolic functions such as the synthesis of NEAA or lipids (see review, Lapierre et al., 2012).

Threonine. As already mentioned for Lys, Thr has also been classified in group 2 AA because of a $U:O$ greater than 1 (Mephram, 1982; Lapierre et al., 2012), as was also observed in the current study. Threonine is unique among group 2 AA because it is metabolized by the PDV, the liver, and the mammary gland but not by the peripheral tissues. Indeed, the proportions metabolized in excess of $NASEM$ -defined export purpose are 0.34, 0.57, and 0.08 for PDV, liver, and mammary gland, respectively, and nonsignificant for PERI, on average across the dataset. So $>90\%$ of potential catabolism is across the splanchnic tissues, with the greater response in removal to increased Thr supply occurring through liver metabolism. Although

considered a group 2 AA, the $U:O$ ratio was lower than for the BCAA and Lys and, furthermore, the ratio did not increase in response to elevated supply, again in contrast to the other group 2 AA. The fact that the $U:O$ ratio of Thr was higher than 1 but lower than the BCAA and Lys and was not altered with Thr supply has already been reported (Lapierre et al., 2012).

A final point of interest is that the PDV_{CATA} of cows in early lactation was greater compared with those in later lactation, with the difference being significant or a tendency for 6 out of the 9 EAA studied. The net removal of EAA not used for MFP but rather used to support the growth of the PDV during the postpartum period (Gibb et al., 1992; Reynolds et al., 2004) could account for part of this EAA capture by the PDV. The potential PDV_{CATA} estimated with Equation [6] would indicate a PDV gain of TP ranging from 3 to 6 kg for the first 30 DIM, using the empty body composition of $NASEM$ (2021); again, these data are associated with very high SD and should be interpreted cautiously. Indeed, the CP gain of the gut obtained from slaughter were much lower, averaging between 0.3 and 0.4 kg between wk 0 and 5 or for the first 22 DIM, respectively (Gibb et al., 1992; Reynolds et al., 2004). Furthermore, it cannot be ruled out that during this period, MFP might be underestimated because of the rapid increment in DMI, which would also result in a higher EAA removal than with the current calculations in $NASEM$ (2021).

CONCLUSIONS

Despite the technical challenge in measuring transorgan net fluxes of EAA and the theoretical concepts underlying the calculation of $NASEM_{EffU}$, the $Measured_{EffU}$ using EAA removal by the different tissues not contributing to protein secretions agreed well with $NASEM_{EffU}$ when cows were sampled at $DIM > 30$, with the CCC averaging 68% across EAA, excluding Trp. These good relationships were based on the premises that the EAA removal by the different tissues and not used to support export protein (or accretion if present) represents their potential catabolism. This catabolism occurred (1) in the liver for group 1 AA (i.e., His, Met, and Phe), (2) across the PDV, PERI, and mammary gland for the BCAA, (3) across the liver, PERI, and mammary gland for Lys, and (4) across the PDV, liver, and mammary gland for Thr. Only Met did not fit as well with these concepts: $Measured_{EffU}$ was smaller than $NASEM_{EffU}$ and thus suggested that the liver was using part of Met removal for the synthesis of metabolites not including other EAA and not related to export proteins. This might suggest that the target $EffU$ of Met proposed by $NASEM$ (2021) should be reduced. For cows sampled at $DIM \leq 30$, the endogenous supply (mobilization) was found to be a significant contributor to mEAA but diffi-

cult to estimate with accuracy. As a result, unsatisfactory relationships between $\text{NASEM}_{\text{EffU}}$ and $\text{Measured}_{\text{EffU}}$ were obtained for most of the EAA. In its current form, the concept of $\text{NASEM}_{\text{EffU}}$ does not apply to cows in early lactation. The sites of EAA removal, however, remained the same in the 2 stages of lactation, with the exception that PERI was a source of endogenous supply for cows in early lactation, but a site of BCAA and Lys catabolism for cows in later lactation.

NOTES

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Nonstandard abbreviations used: 3MH = 3-methyl-His; BCAA = branched-chain AA; CCC = concordance correlation coefficient; ECT = error in central tendency; Eff = efficiency; EffU = efficiency of utilization; EndoSupply = endogenous supply; EndoUri = AA in endogenous urinary losses; ER = error due to deviation of the regression slope from unity; HEP_{CATA} = hepatic catabolism; HEP_{NF} = hepatic net flux; HMBi = 2-hydroxy-4-(methylthio)-butanoic acid isopropyl ester; IQR = interquartile range; Max. = maximum; mEAA = metabolizable EAA; $\text{Measured}_{\text{EffU}}$ = measured efficiency of utilization; MFP = metabolic fecal protein; MG_{CATA} = catabolism by the mammary gland; MG_{NF} = mammary net flux; Min. = minimum; Mpf = mammary plasma flow; MPY = milk true protein yield; $\text{MilkAA}_{\text{BB}}$ = AA in milk protein secretion from blood-borne proteins; $\text{MilkAA}_{\text{MG}}$ = AA in milk protein secretion synthesized in the mammary gland (i.e., excluding blood-borne proteins); $\text{MilkAA}_{\text{TOT}}$ = EAA in total milk protein secretion; MSPE = mean square prediction error; NASEM = National Academies of Sciences, Engineering, and Medicine; $\text{NASEM}_{\text{EffU}}$ = efficiency of utilization computed by NASEM; obs.mean = average of measured values; pAH = *p*-aminohippuric acid; PDV = portal-drained viscera; PDV_{CATA} = catabolism by the PDV; PDV_{NF} = PDV net flux; PERI = nonmammary peripheral tissues; $\text{PERI}_{\text{CATA}}$ = catabolism by the peripheral tissues; $\text{PostTSP}_{\text{CATA}}$ = catabolism by the postsplanchnic tissues; RMSPE = square root of mean square prediction error; RPE = relative prediction error; RSR = ratio of square root of mean

square prediction error to SD of measured data; std = standardized; TP = true protein; TSP_{NF} = splanchnic tissues net flux; U:O = ratio of mammary AA uptake to AA in milk protein secretion synthesized in the mammary gland; VAdiff = venous-arterial difference.

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